Supporting Information

Chemically diverse antimicrobial peptides induce hyperpolarization of the *E. coli* membrane

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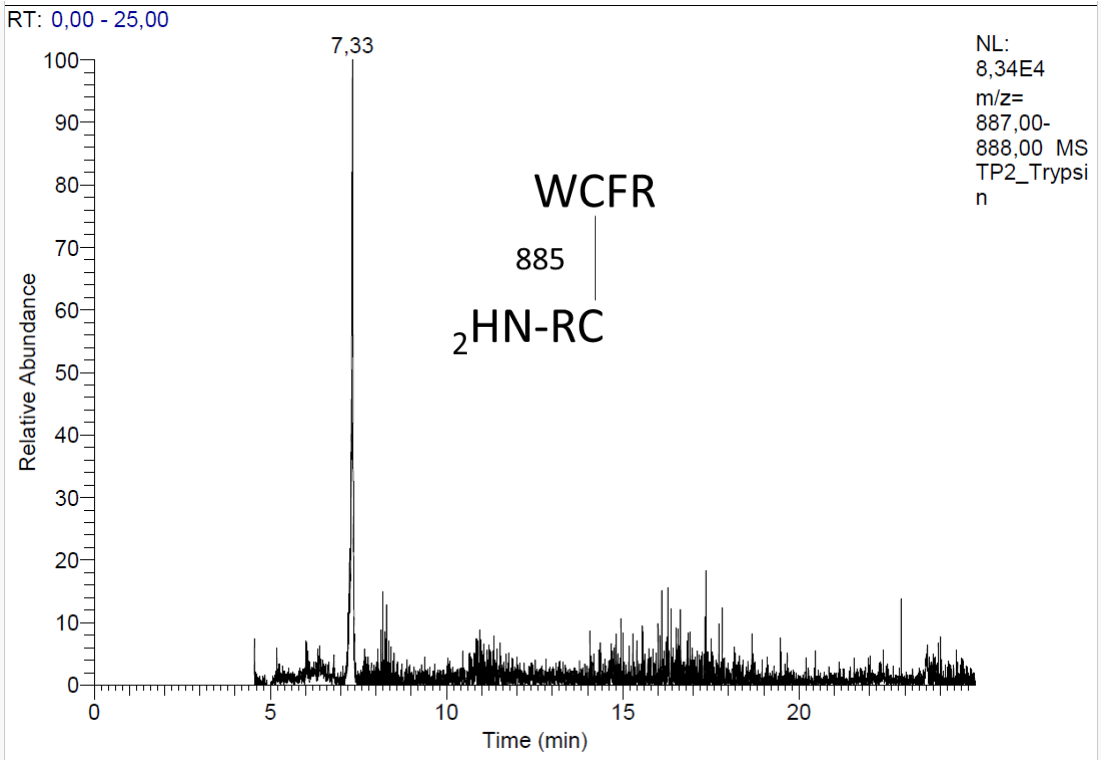
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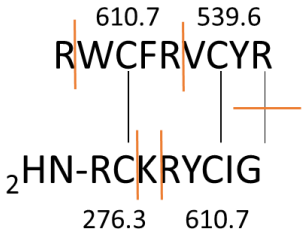
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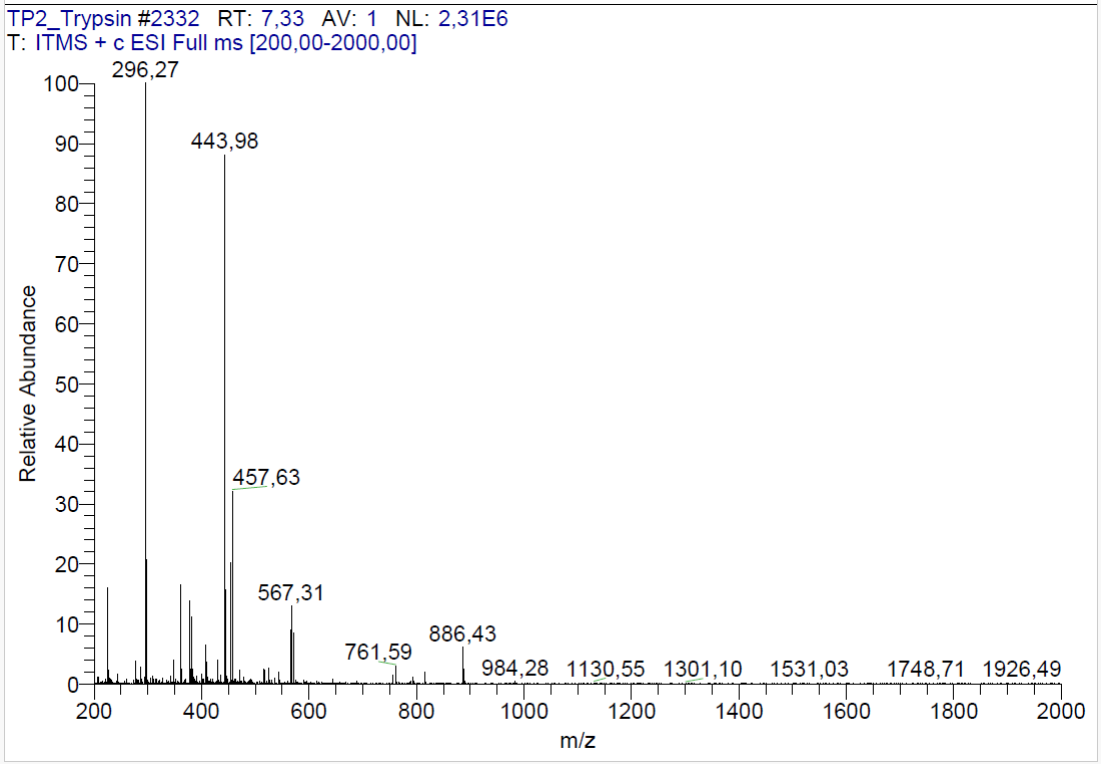
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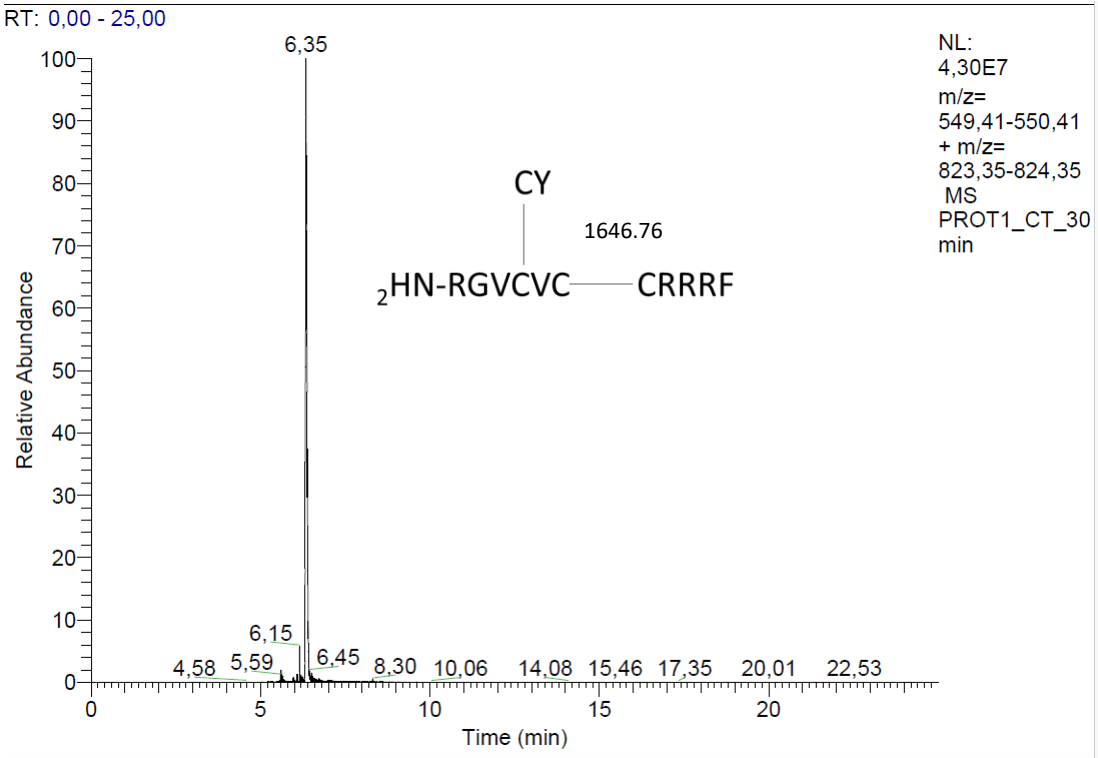
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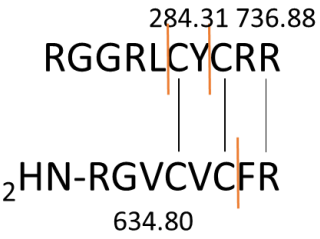


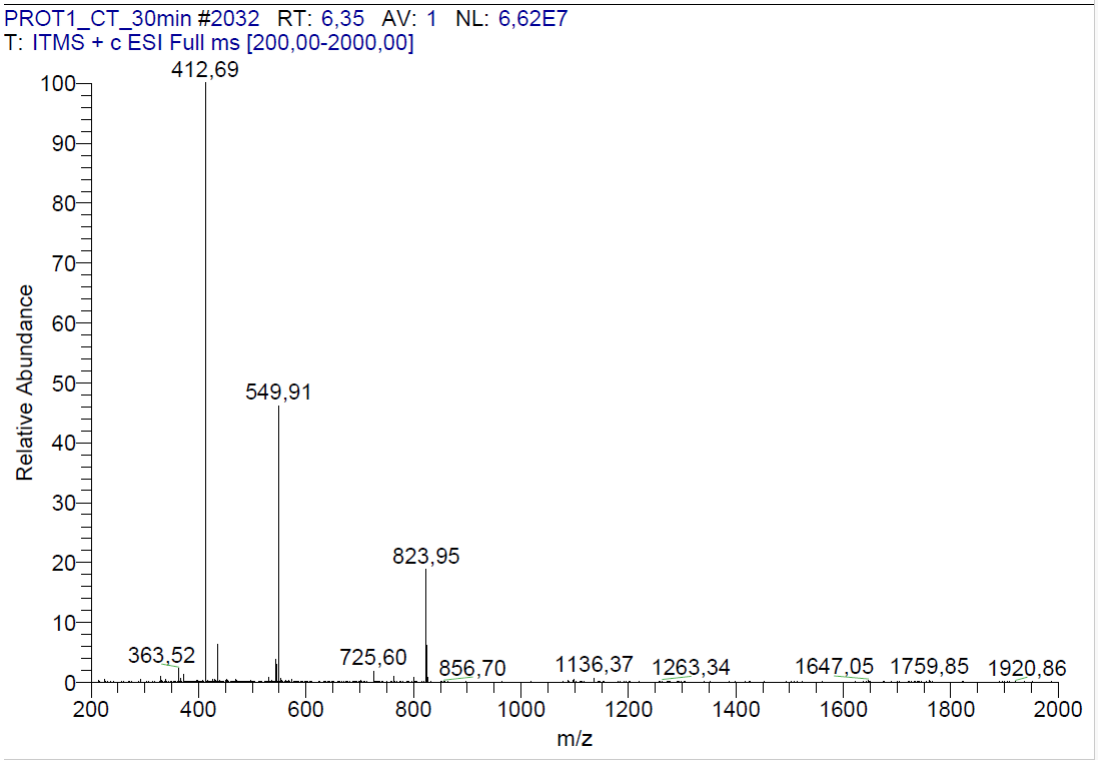




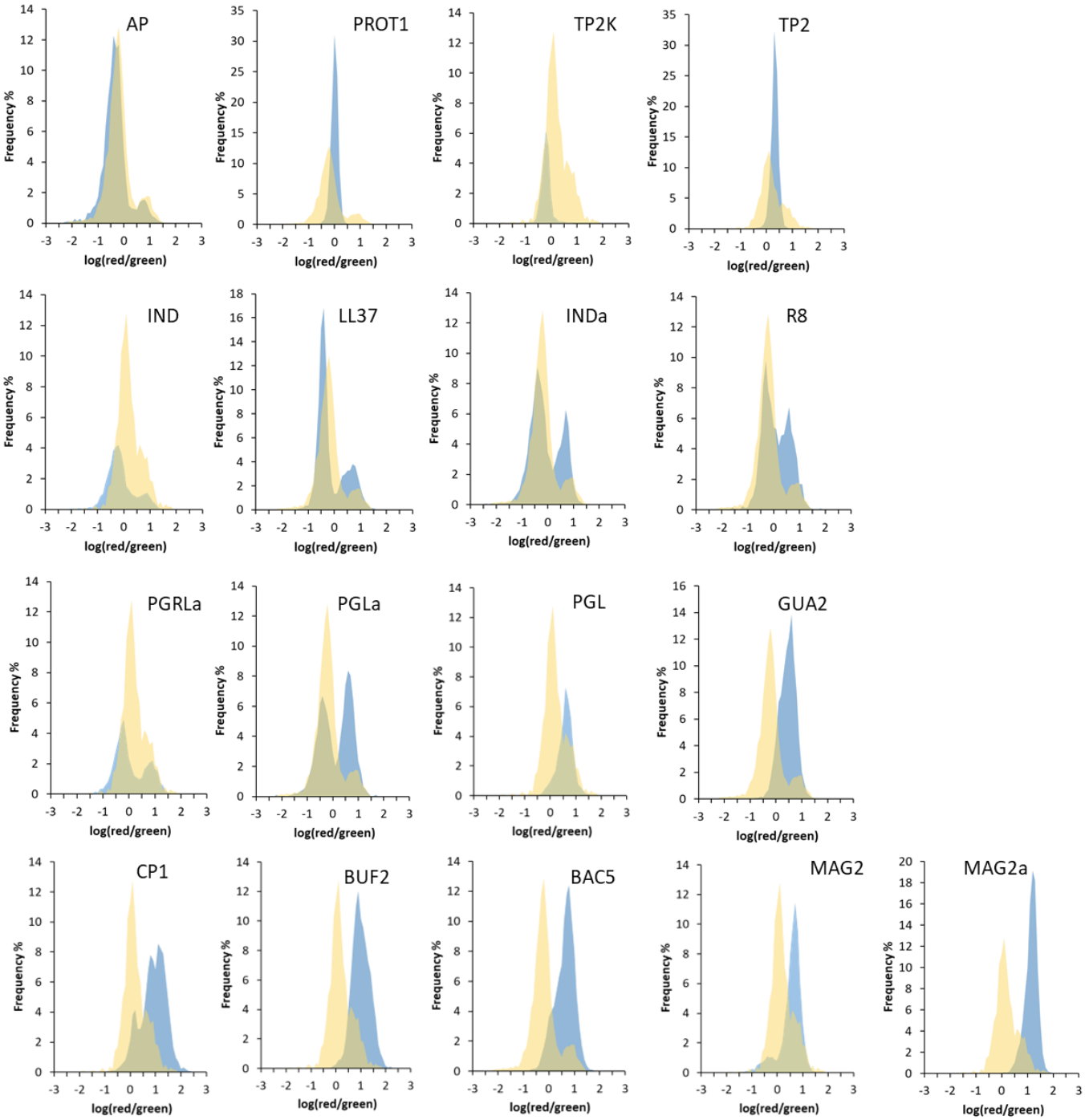
Supplementary Figure 1. Selective HPLC-MS trace of the TP2 fragment obtained after tryptic digestion and its mass spectrum recorded at 7.33 min. Expected MW of the fragment: 885.08. Expected m/z values: $[M+H]^{+}$: 886.08; $[M+2H]^{2+}$: 443.53; $[M+3H]^{3+}$: 296.02. The cleavage sites of the digestion are indicated by red lines. No fragment was observed from any other possible disulfide bond combinations.



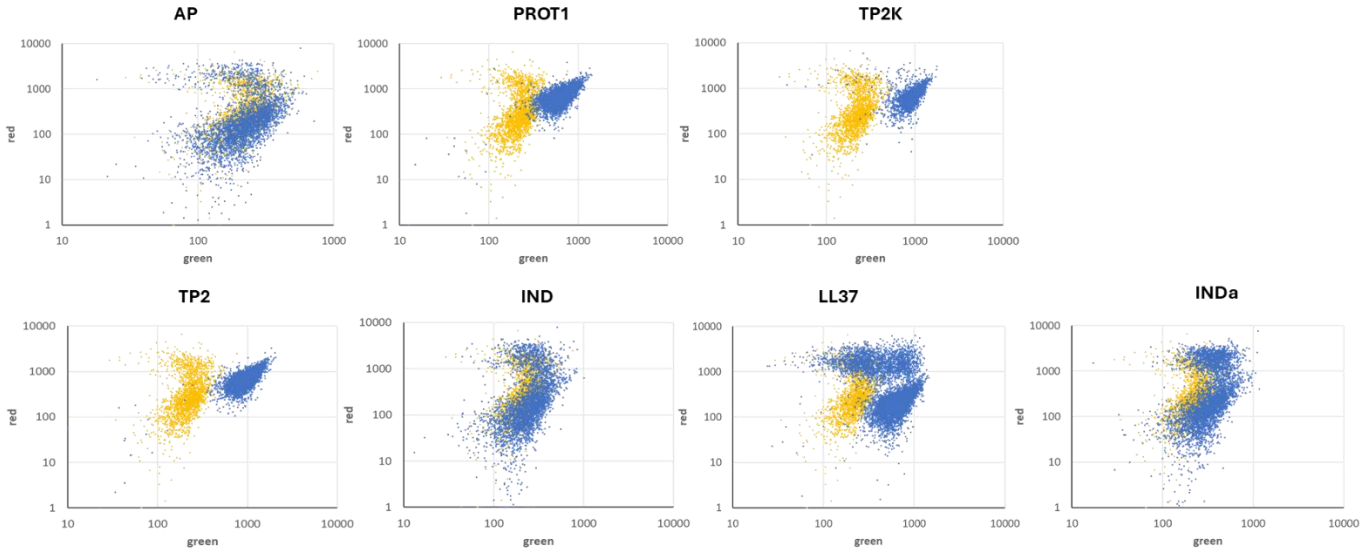




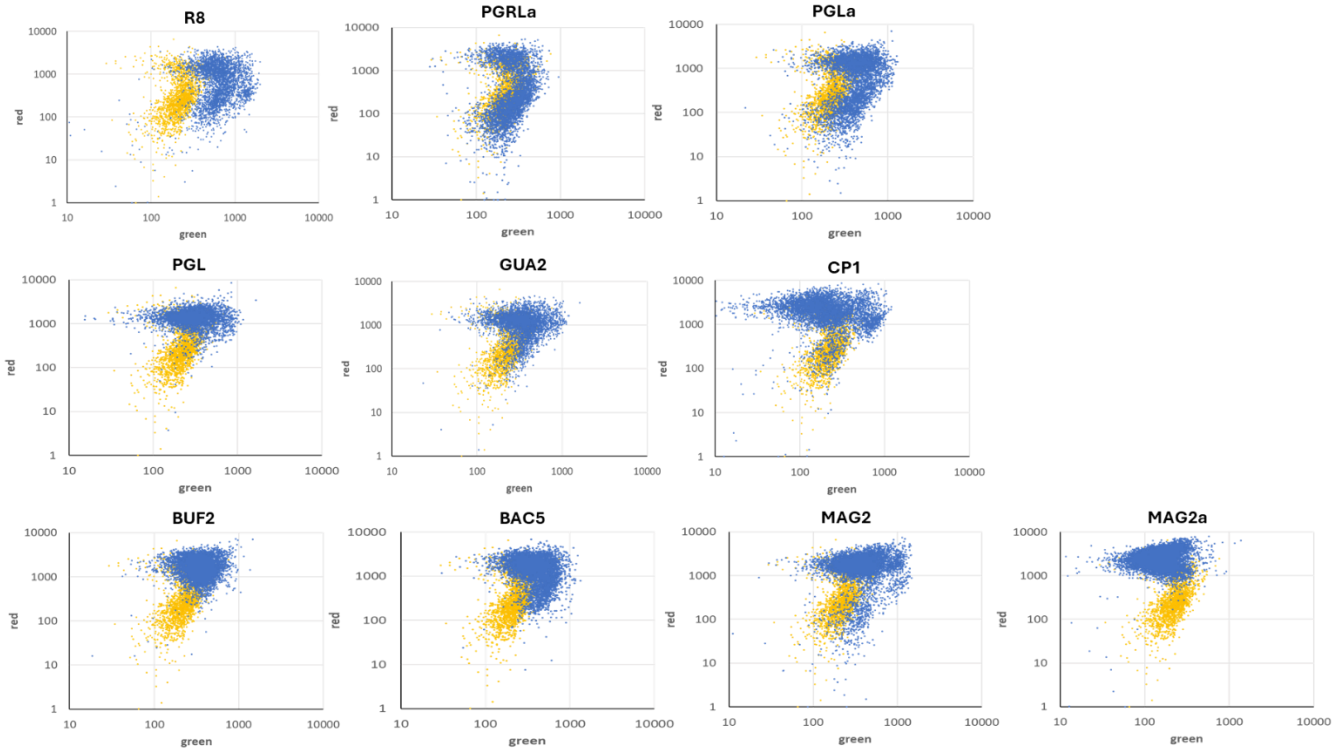
Supplementary Figure 2. Selective HPLC-MS trace of the PROT1 fragment obtained after chymotryptic digestion, and its mass spectrum recorded at 6.35 min. Expected MW of the fragment: 1646.76. Expected m/z values: $[M+2H]^{2+}$: 824.38; $[M+3H]^{3+}$: 549.92 $[M+4H]^{4+}$: 412.69. No fragment was observed from any other possible disulfide bond combinations.



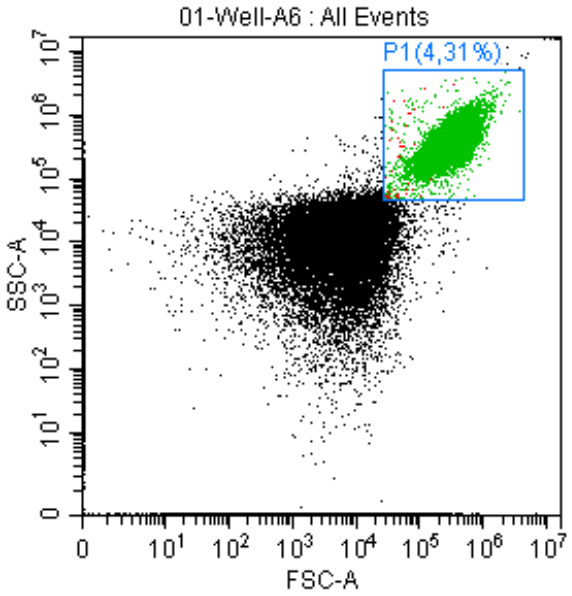
Supplementary Figure 3. Bacterial hyperpolarization effects of AMPs at subinhibitory concentrations. Flow cytometric membrane polarization assay (BacLight) was carried out in *E. coli* using fluorescent membrane potential indicator dye $DiOC_2(3)$. Histograms of the log intensity ratio calculated from the red and green channels for AP, PROT1, TP2K, TP2, IND, LL37, INDa, R8, PGRLa, PGLa, PGL, GUA2, CP1, BUF2, BAC5, MAG2 and MAG2a.



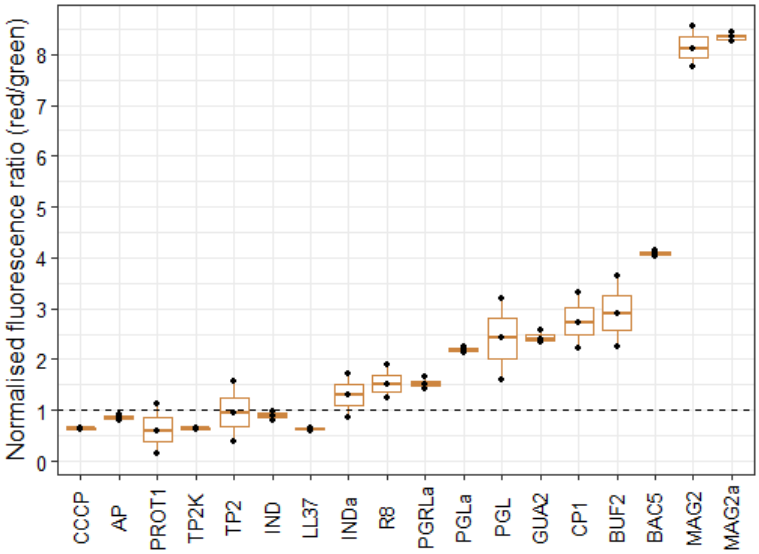
Supplementary Figure 4. Membrane polarization effects of AMPs with non-membrane-inducible secondary structures at sub-inhibitory concentrations. Flow cytometric membrane polarisation assay (BacLight) was carried out in *E. coli* using fluorescent membrane potential indicator dye $DiOC_2(3)$. Flow cytometry density plot of the red and green channels measured for membrane-insensitive AMPs. Yellow-untreated, blue – treated.



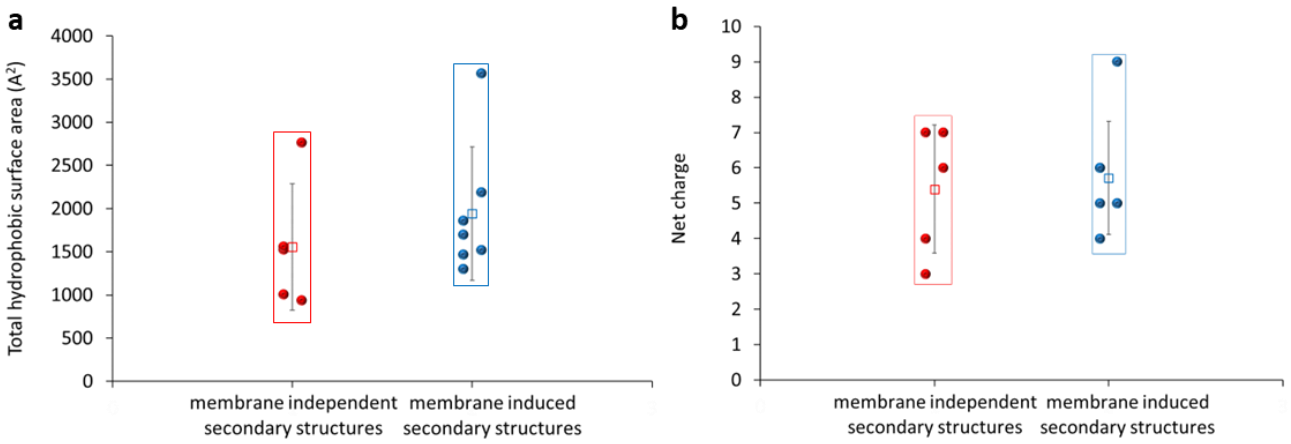
Supplementary Figure 5. Membrane polarization effects of AMPs with membrane-inducible secondary structures at subinhibitory concentrations. Flow cytometric membrane polarisation assay (BacLight) was carried out in *E. coli* using fluorescent membrane potential indicator dye $DiOC_2(3)$. Flow cytometry density plot of the red and green channels measured for membrane-sensitive AMPs. Yellow-untreated, blue – treated.



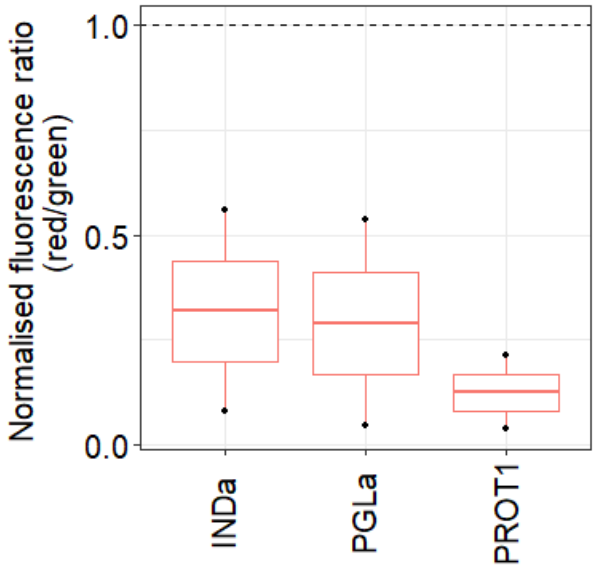
Supplementary Figure 6. Gating strategy used for flow cytometry analysis. FACS plots are shown as representative example of the gating strategy used in each experiment. The gating strategy was based on the forward scatter (FSC-A) and side scatter (SSC-A) values both above $10e^5$.



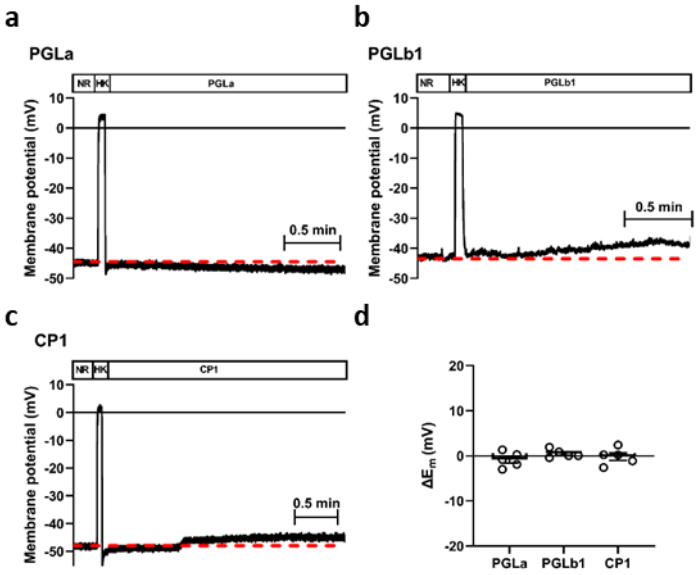
Supplementary Figure 7. Membrane polarization effect of AMPs at subinhibitory concentrations. The normalized red/green fluorescence ratio was calculated using population mean fluorescence intensities for cells treated with and without the AMPs. A ratio above one indicates hyperpolarization. Data are based on three biological replicates.



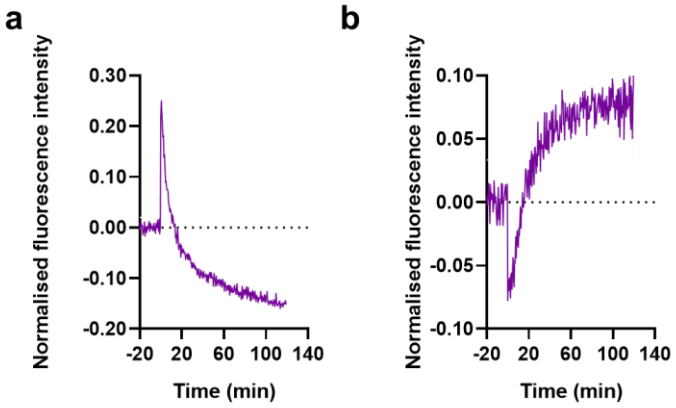
Supplementary Figure 8. The effects of the sequence-dependent hydrophobicity and electrostatic parameters on the membrane inducibility. Neither the total hydrophobic surface area (**a**) nor the net charge (**b**) displays significant influence on the membrane inducibility, indicating the necessity of a complex and dynamic structural behaviour.



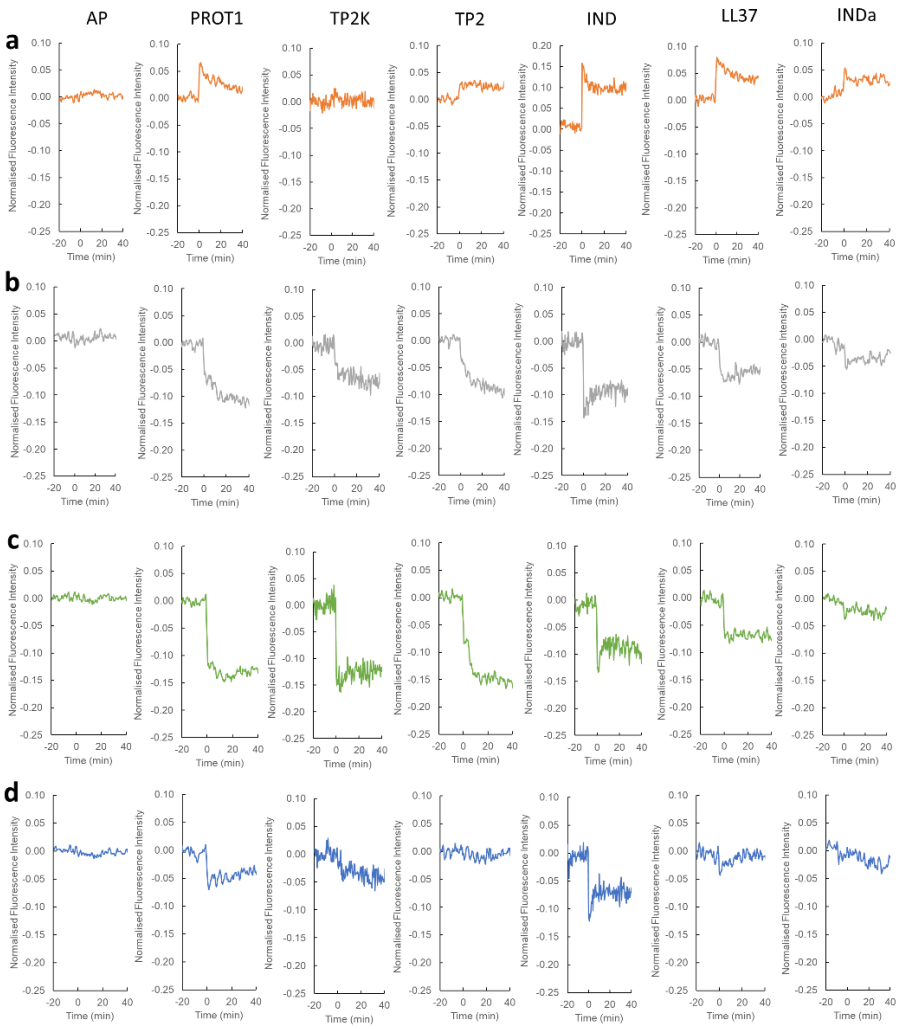
Supplementary Figure 9. Membrane depolarization effects in *S. epidermidis* observed for AMPs. MIC values were determined for AMPs, and sequences with acceptable MICs were tested for membrane polarization effects at 0.5 MIC. Fluorescence values were calculated relative to the untreated control strain. A decrease in the ratio corresponds to depolarization, whereas an increase in the ratio corresponds to a hyperpolarised membrane potential. CCCP use as a control because it eradicates the proton gradient, eliminating the bacterial membrane potential.



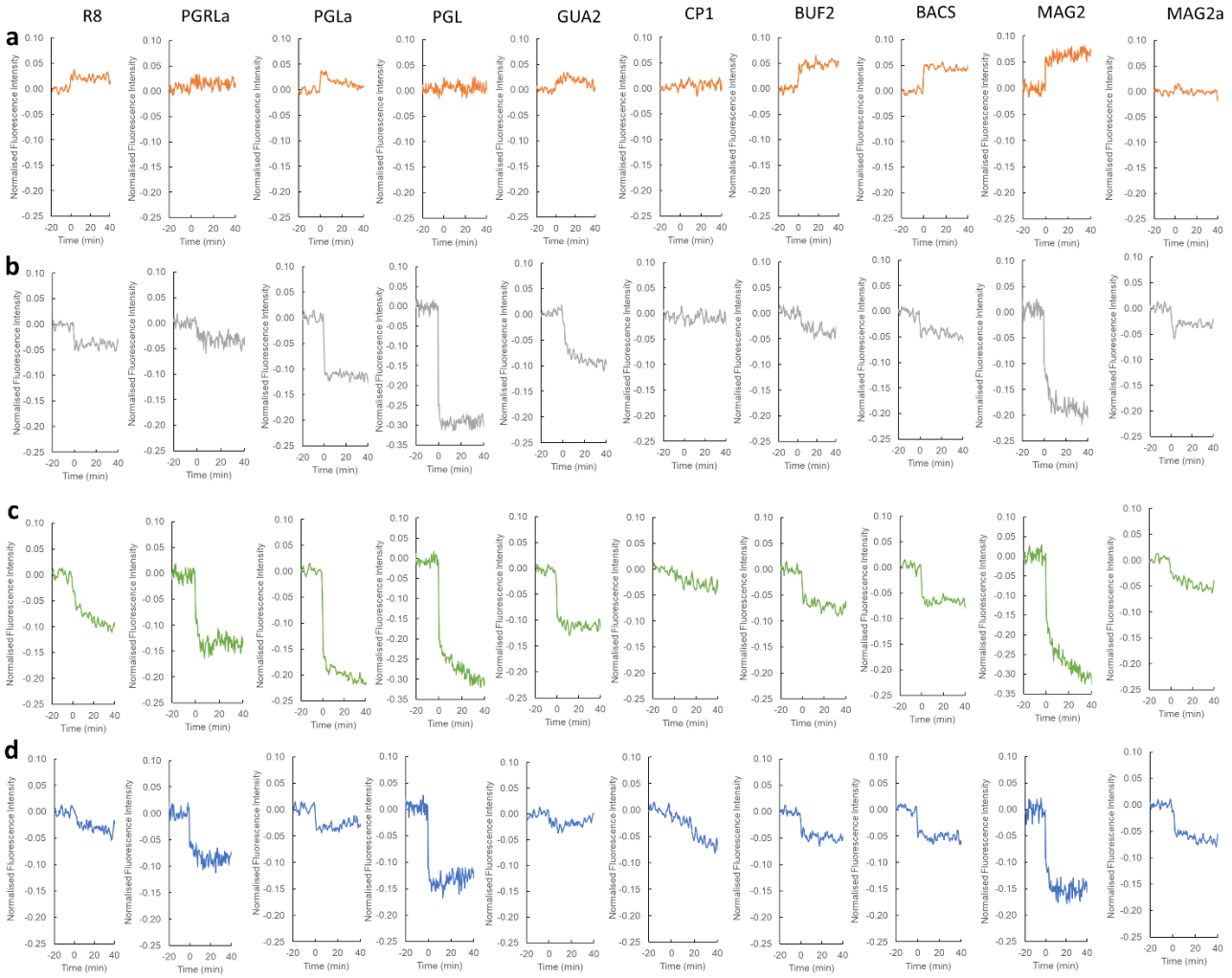
Supplementary Figure 10. Patch-clamp membrane potential measurements for AMPs in human peripheral lymphocytes. Effects of PGLa, PGLb1, and CP1 exposition on the membrane potential of activated human peripheral lymphocytes were investigated by using the patch-clamp technique in the current-clamp mode. First, cells were perfused with a sodium-based control bath solution (normal ringer, NR); then as a positive control, a solution with high potassium concentration (HK) was used, which was followed by the application of PGLa-, PGLb1- or CP1-containing solutions (**a**, **b** and **c**, respectively). All the materials were applied at 0.5 MIC and dissolved in NR. As is also seen in representative traces (**a**-**c**) and column graph (**d**), none of the examined materials caused biologically relevant changes in the resting membrane potential. Data are shown as mean ± SEM (n=5), and dots represent values obtained from individual cells.



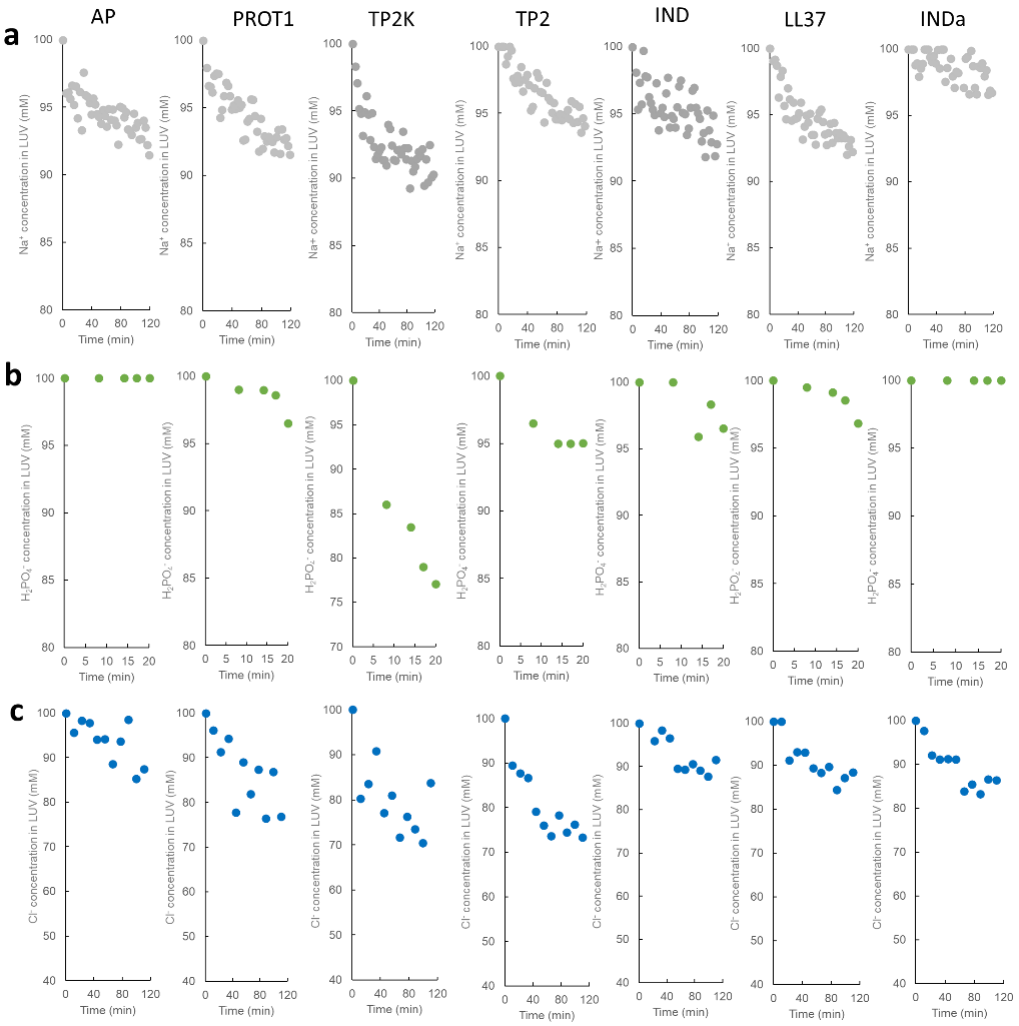
Supplementary Figure 11. Validation of the LUV-based membrane polarization model with the K^+ selective ionophore valinomycin (VAL) and ion gradient inversion. (**a**) The relative fluorescence level changes of the membrane potential-dependent dye oxonol VI with 100 mM NaCl in the intra- and 100 mM KCl in the extravesicular space The inside positive membrane polarization is indicated by negative changes after a transient period after the peptide injection. (**b**) The relative fluorescence level changes of the membrane potential-dependent dye oxonol VI with 100 mM KCl in the intra- and 100 mM NaCl in the extravesicular space. The inside negative membrane polarization is indicated by positive changes after a transient period after the peptide injection.



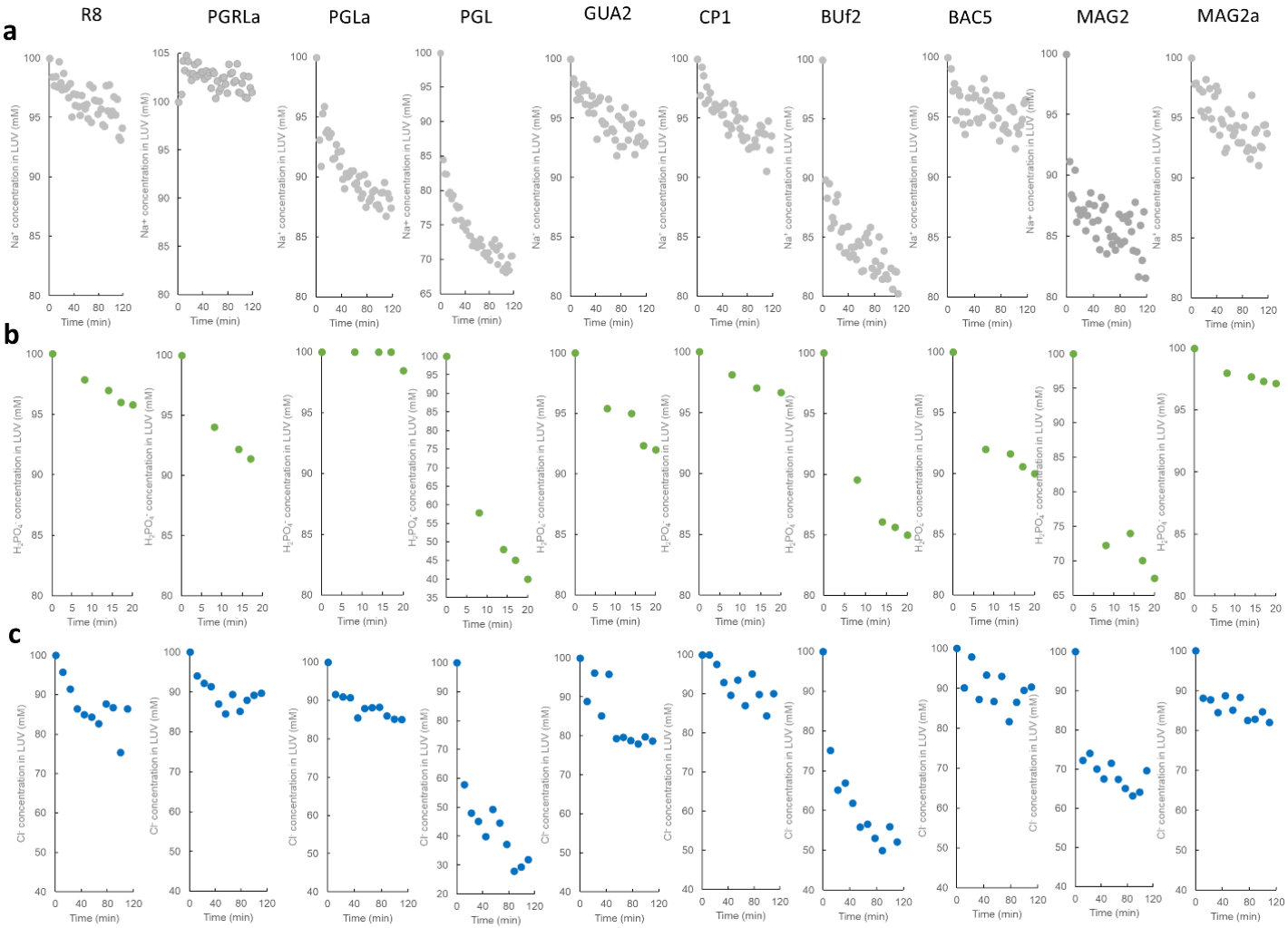
Supplementary Figure 12. AMP-induced membrane polarization in the LUV model using AMPs with non-membrane-inducible secondary structures. Peptide-induced membrane potentials upon adding AP, PROT1, TP2K, TP2, IND, LL37 and INDa. The relative fluorescence level change of the membrane potential-dependent dye oxonol VI was monitored. A negative change indicates a positive inside potential (see Supplementary Figure 8 for validation). Experiments without ion gradient: 100 mM NaCl in the intra- and the extravesicular space (orange curves) (**a**). Experiments with Na^+/K^+ ion gradient: 100 mM NaCl in the intra- and 100 mM KCl in the extravesicular space (grey curves) (**b**). Experiments with Cl^-/ $H_2PO_4^-$ ion gradient: 100 mM NaCl in the intra- and 100 mM NaH_2PO_4 in the extravesicular space (green curves) (**c**). Experiments with Cl^-/NO_3^- ion gradient: 100 mM NaCl in the intra- and 100 mM $NaNO_3$ in the extravesicular space (blue curves) (**d**).



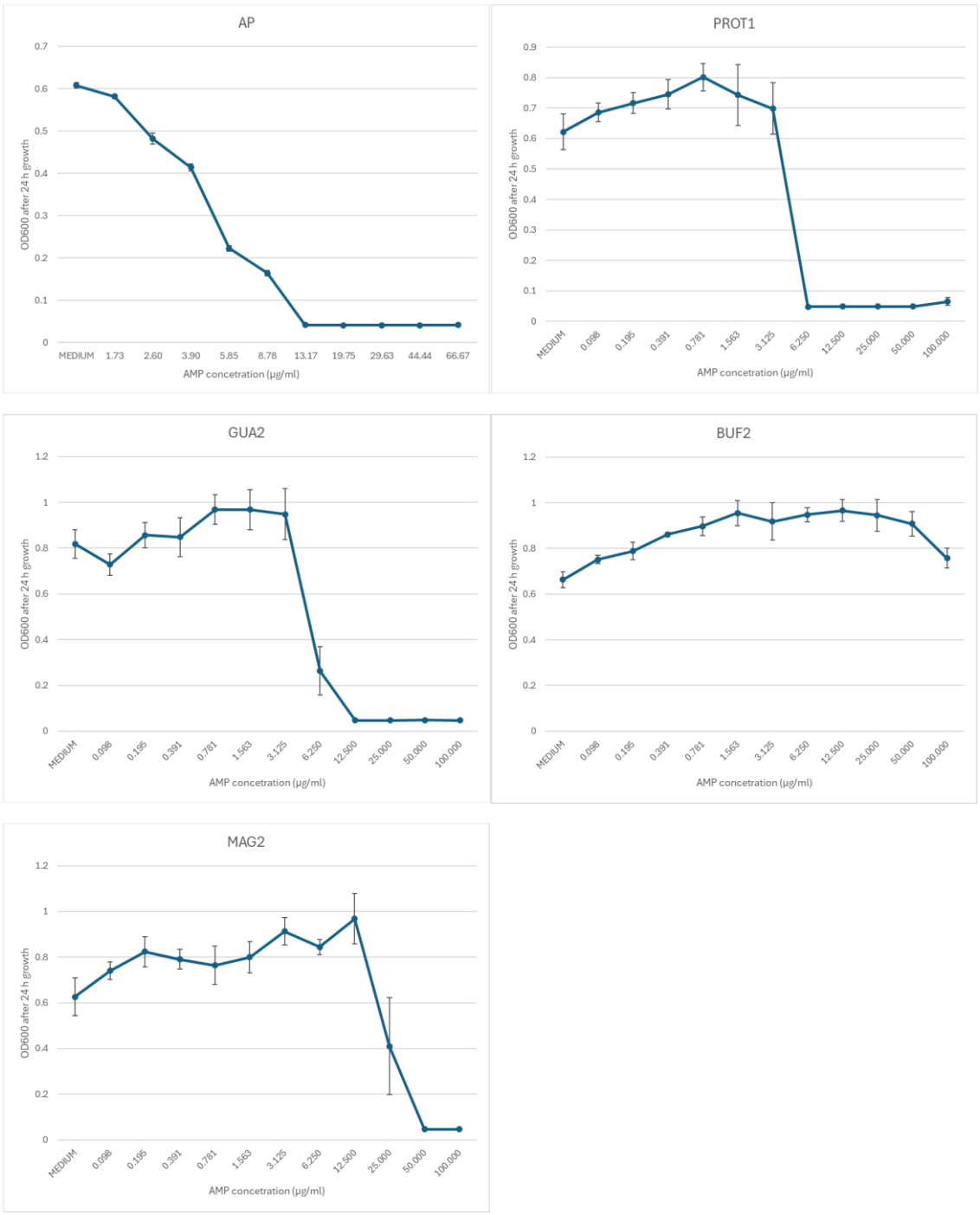
Supplementary Figure 13. AMP-induced membrane polarization in the LUV model using AMPs with membrane-inducible secondary structures. Peptide-induced membrane potentials upon adding R8, PGRLa, PGLa, PGL, GUA2, CP1, BUF2, BAC5, MAG2 and MAG2a. The relative fluorescence level change of the membrane potential-dependent dye oxonol VI was monitored. A negative change indicates a positive inside potential (see Supplementary Figure 8 for validation). Experiments without ion gradient: 100 mM NaCl in the intra- and the extravesicular space (orange curves) (**a**). Experiments with Na^+/K^+ ion gradient: 100 mM NaCl in the intra- and 100 mM KCl in the extravesicular space (grey curves) (**b**). Experiments with $Cl^-/ H_2PO_4^-$ ion gradient: 100 mM NaCl in the intra- and 100 mM NaH_2PO_4 in the extravesicular space (green curves) (**c**). Experiments with Cl^-/NO_3^- ion gradient: 100 mM NaCl in the intra- and 100 mM $NaNO_3$ in the extravesicular space (blue curves) (**d**).



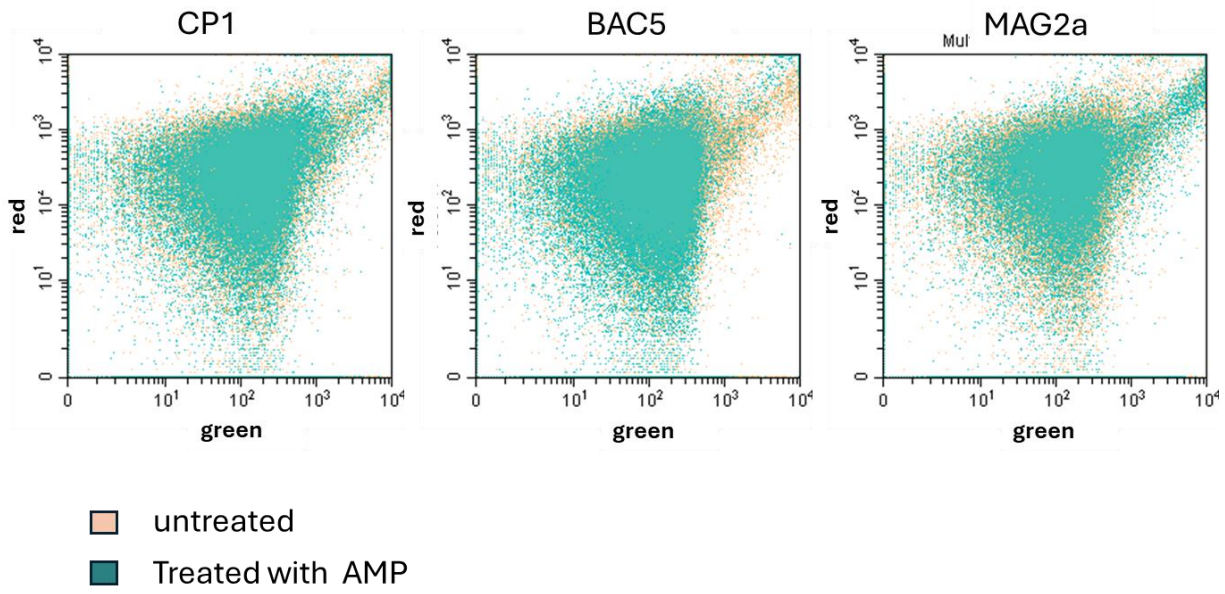
Supplementary Figure 14. Ion exchange measurements in the LUV model by NMR using AMPs with non-membrane-inducible secondary structures. Ion concentrations determined by ^{23}Na, ^{31}P and ^{35}Cl NMR assays upon adding AP, PROT1, TP2K, TP2, IND, LL37 and INDa. External and internal ion signals were distinguished with NMR chemical shift reagents 1 mM $Dy(PPP)_2^{7-}$ and 4 mM $Co(NO_3)_2$ for ^{23}Na and ^{35}Cl, respectively. No shift reagent was needed for ^{31}P because the inside and outside signals were distinguishable. Intravesicular Na^+ concentration changes (grey) in the Na^+/K^+ exchange experiment (100 mM NaCl in the intra- and 100 mM KCl in the extravesicular space) (**a**). Intravesicular $H_2PO_4^-$ concentration changes (green) in the $Cl^-/H_2PO_4^-$ exchange experiments (100 mM NaCl in the extra- and 100 mM NaH_2PO_4 in the intravesicular space) (**b**). Intravesicular Cl^- concentration changes (blue) in the Cl^-/NO_3^- exchange experiments (100 mM NaCl in the intra- and 100 mM $NaNO_3$ in the extravesicular space) (**c**).



Supplementary Figure 15. Ion exchange measurements in the LUV model by NMR using AMPs with membrane-inducible secondary structures. Ion concentrations determined by ^{23}Na, ^{31}P and ^{35}Cl NMR assays upon adding R8, PGRLa, PGLa, PGL, GUA2, CP1, BUF2, BAC5, MAG2 and MAG2a. External and internal ion signals were distinguished with NMR chemical shift reagents 1 mM $Dy(PPP)_2^{7-}$ and 4 mM $Co(NO_3)_2$ for ^{23}Na and ^{35}Cl, respectively. No shift reagent was needed for ^{31}P because the inside and outside signals were distinguishable. Intravesicular Na^+ concentration changes (grey) in the Na^+/K^+ exchange experiment (100 mM NaCl in the intra- and 100 mM KCl in the extravesicular space) (**a**). Intravesicular $H_2PO_4^-$ concentration changes (green) in the $Cl^-/H_2PO_4^-$ exchange experiments (100 mM NaCl in the extra- and 100 mM NaH_2PO_4 in the intravesicular space) (**b**). Intravesicular Cl^- concentration changes (blue) in the Cl^-/NO_3^- exchange experiments (100 mM NaCl in the intra- and 100 mM $NaNO_3$ in the extravesicular space) (**c**).



Supplementary Figure 16. Dose-response curves of *E. coli* K-12 BW25113 strain for selected AMPs. For abbreviations, see Table 1. Each data point shows the mean ± s.e.m. of two biological replicates.



Supplementary Figure 17. $DiOC_2(3)$ fluorescence distributions in the red and green channels in the dead bacterial population with and without AMPs. The fluorescence response of the cationic dye does not change upon adding AMPs in dead bacteria, indicating the absence of any direct AMP-$DiOC_2(3)$ interaction.

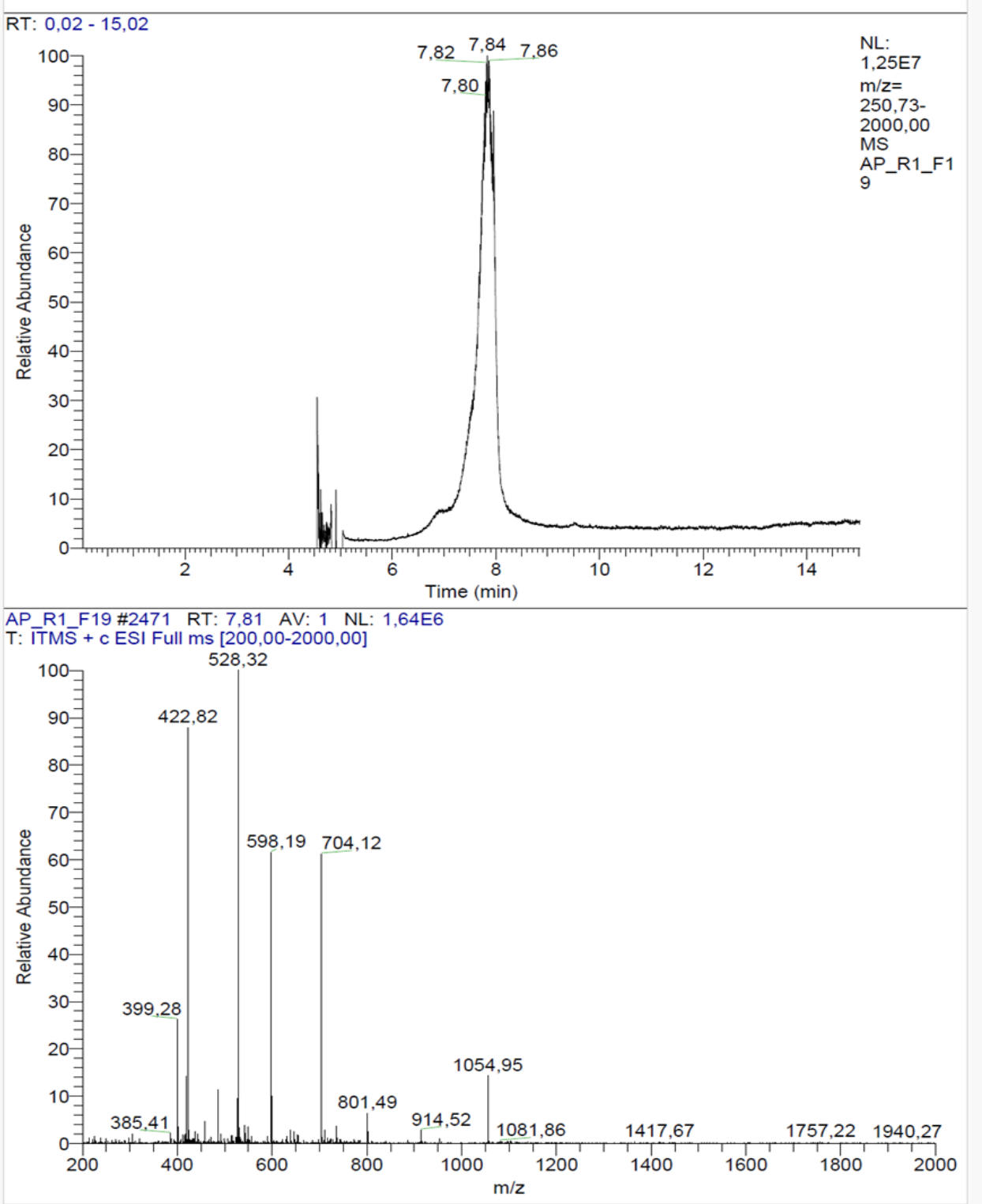
Supplementary Table 1. Minimum inhibitory concentrations of AMPs against *E. coli* BW25113 K-12

Peptides	Abbreviation	MIC (ug/ml)	MIC (μM)
Apidaecin 1B	AP	19.75	9.4
Protegrin 1	PROT1	3.91	1.8
Tachyplesin II - Lys	TP2K	3.91	1.8
Tachyplesin II	TP2	3.91	1.7
Indolicidin-OH	IND	19.75	10.4
LL37	LL37	18.81	4.2
Indolicidin	INDa	8.78	4.6
R8	R8	13.33	6.0
PGLa - Arg	PGRLa	3.91	1.9
PGLa	PGLa	18.1	4.6
PGLa-OH	PGL	90	38.1
Guavanin 2	GUA2	15.68	5.9
Cecropin P1	CP1	2.63	0.8
Buforin II	BUF2	>250	102.7
Bactenecin 5	BAC5	13.33	2.6
Magainin II-OH	MAG2	93.75	38
Magainin II	MAG2a	8.78	3.6

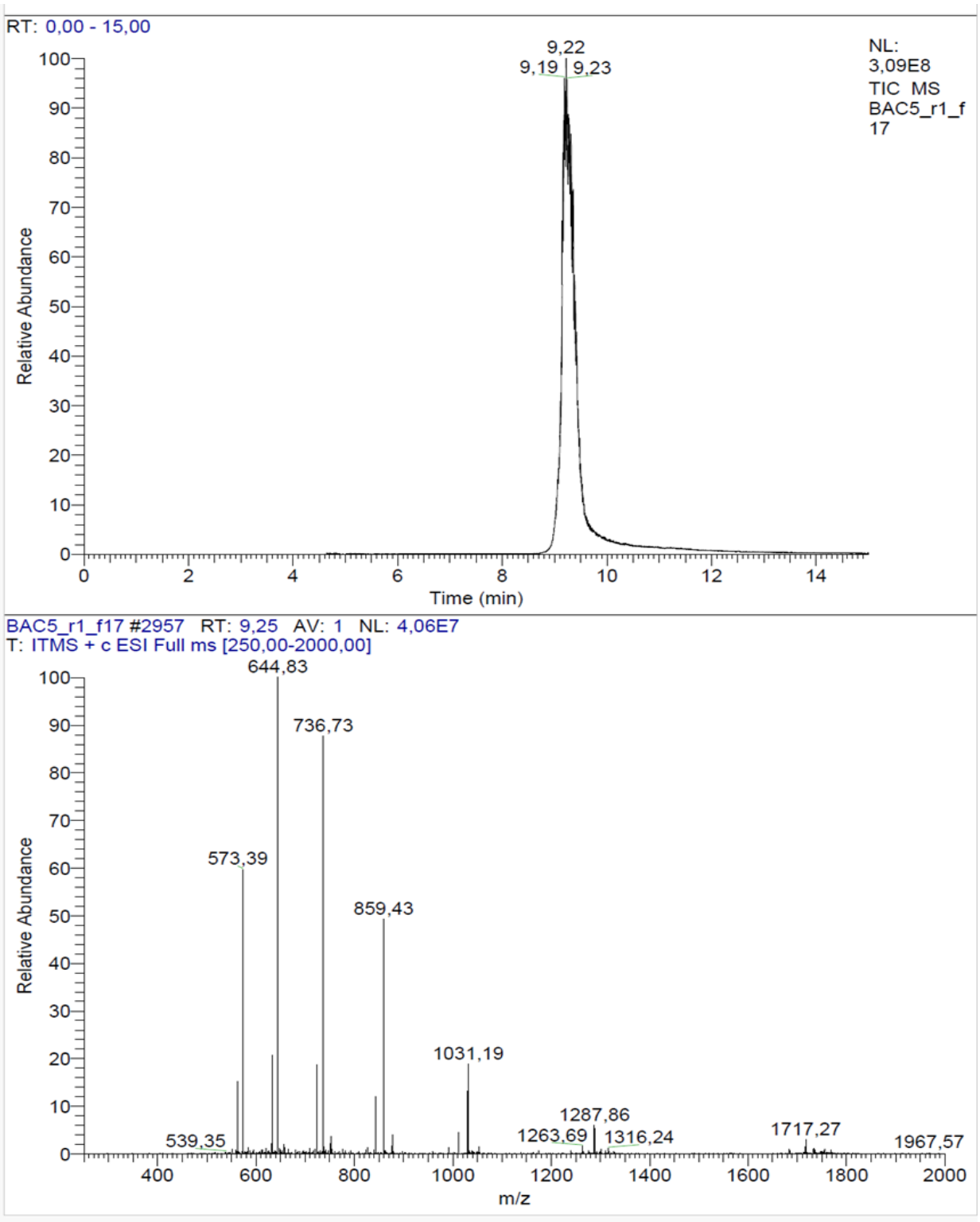
Supplementary Table 2. Minimum inhibitory concentrations of AMPs against *S. epidermidis ATCC 51625*

AMP	*S.epidermidis*		*E.coli*	
	MIC (ug/ml)	MIC (μM)	MIC (ug/ml)	MIC (μM)
AP	>640	303.5	19.75	9.4
PROT1	5	2.3	3.91	1.8
INDa	20	10.5	8.78	4.6
PGLa	40	20.3	18.1	4.6
GUA2	>640	242.1	15.68	5.9
BUF2	>2400	985.7	250	102.7
BAC5	640	124.3	13.33	2.6
MAG2a	160	64.9	8.78	3.6

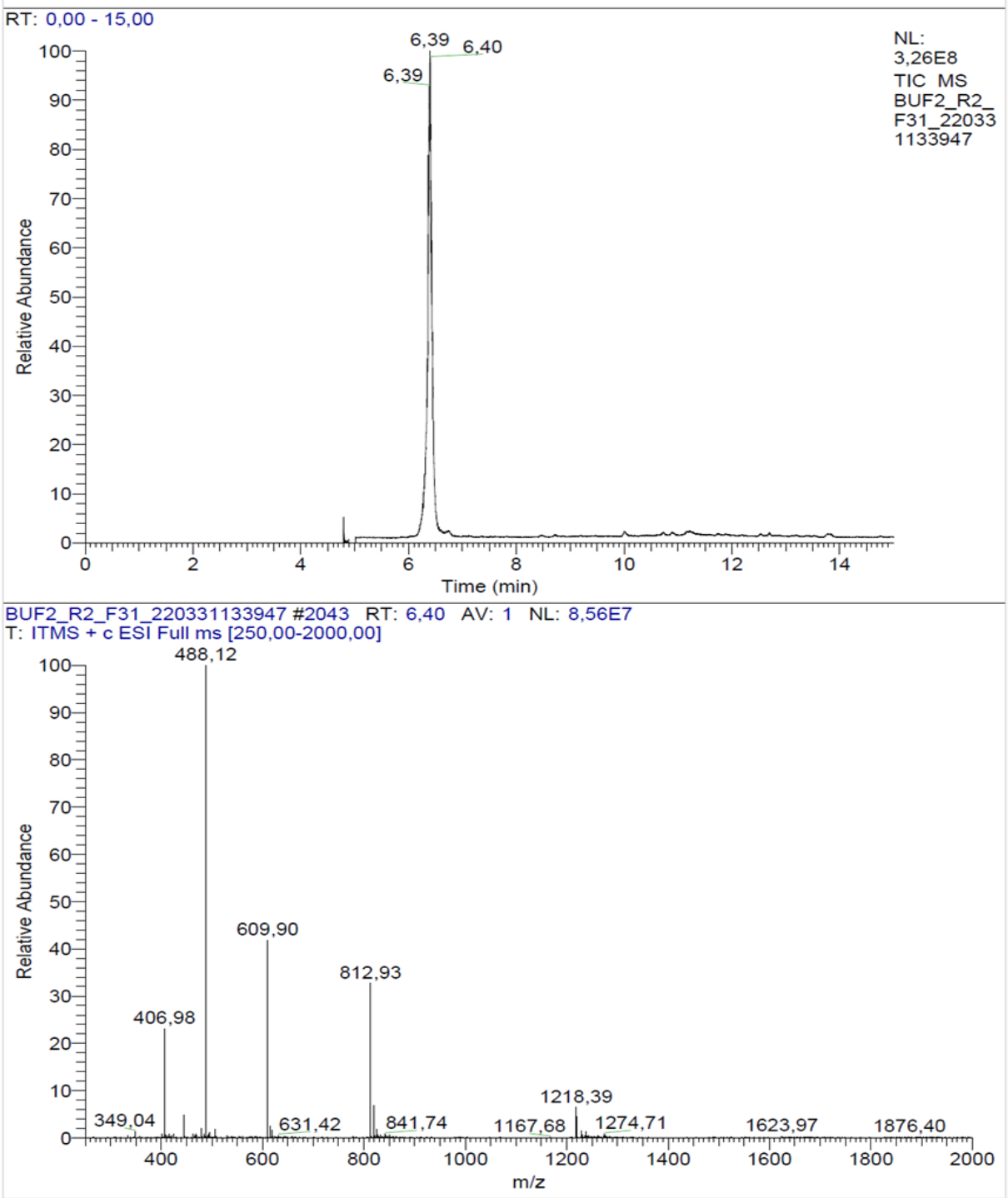
HPLC-MS analytical data



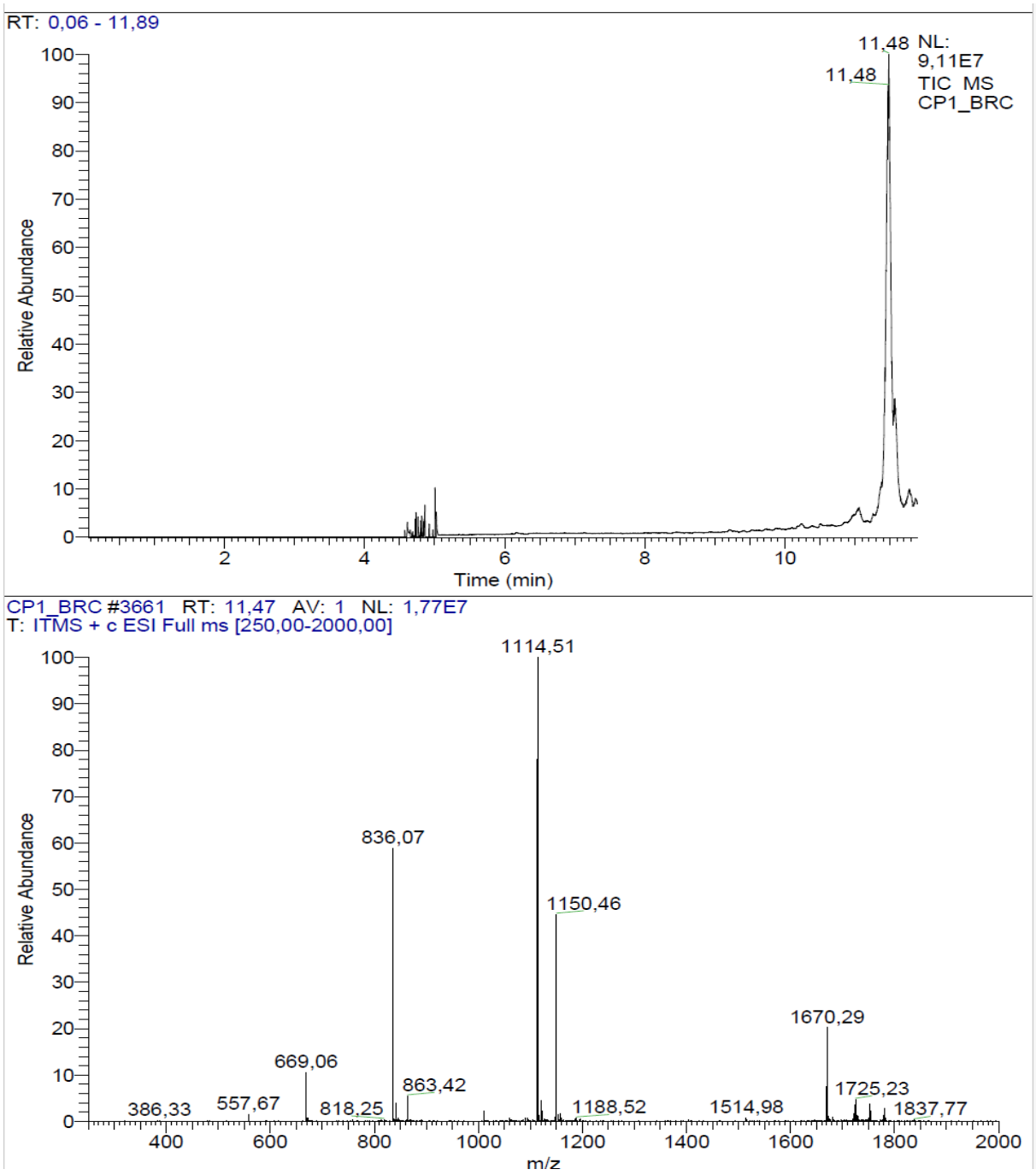
AP. MW: 2108.45. Expected m/z values: $[M+2H]^{2+}$: 1055.23; $[M+3H]^{3+}$: 703.82 $[M+4H]^{4+}$: 528.12



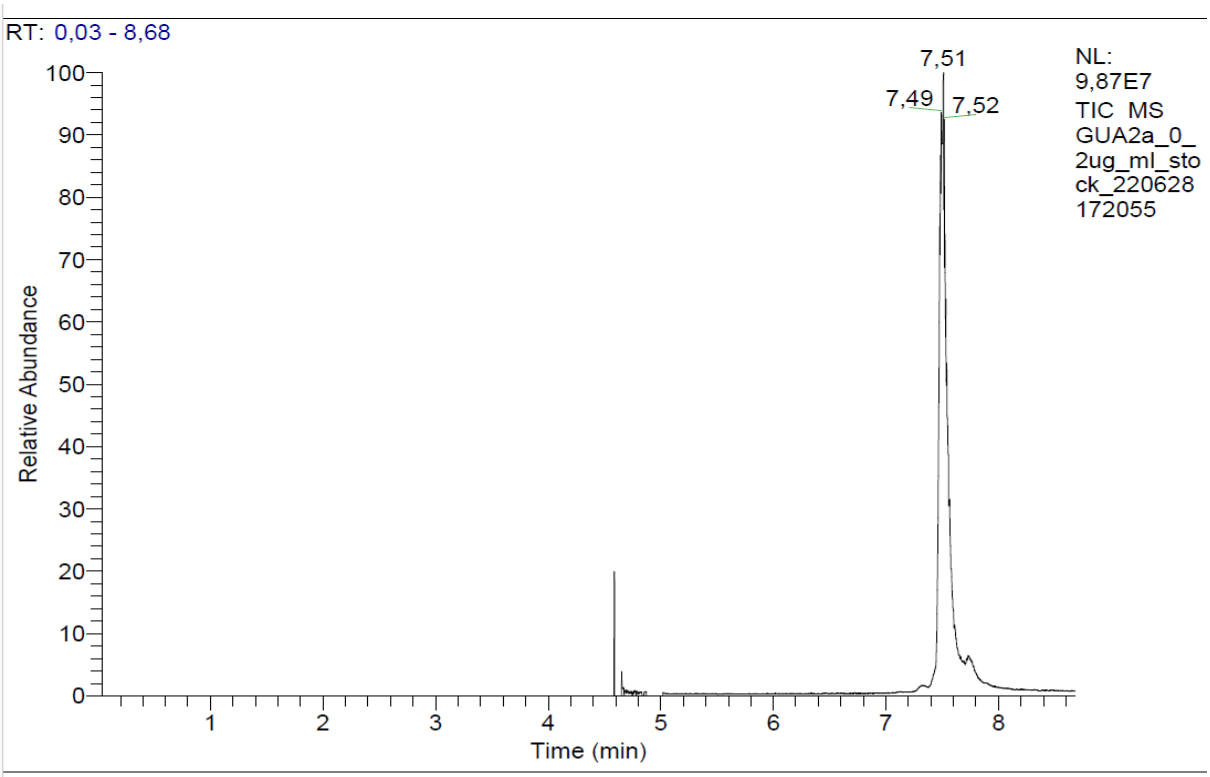
BAC5. MW: 5148.39. Expected m/z values: $[M+3H]^{3+}$: 1717.14 $[M+4H]^{4+}$: 1288.10 $[M+5H]^{5+}$: 1030.69 $[M+6H]^{6+}$: 859.07

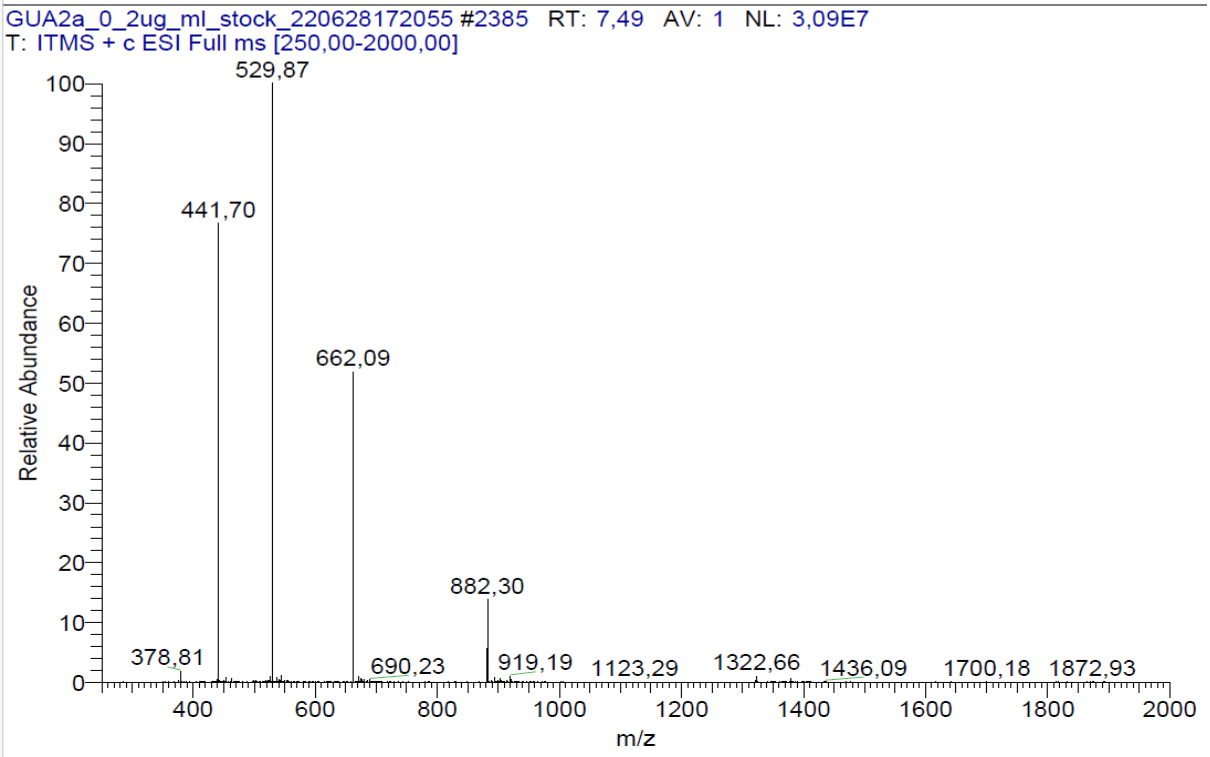


BUF2. MW: 2434.85. Expected m/z values: $[M+2H]^{2+}$: 1218.43; $[M+3H]^{3+}$: 812.62 $[M+4H]^{4+}$: 609.72

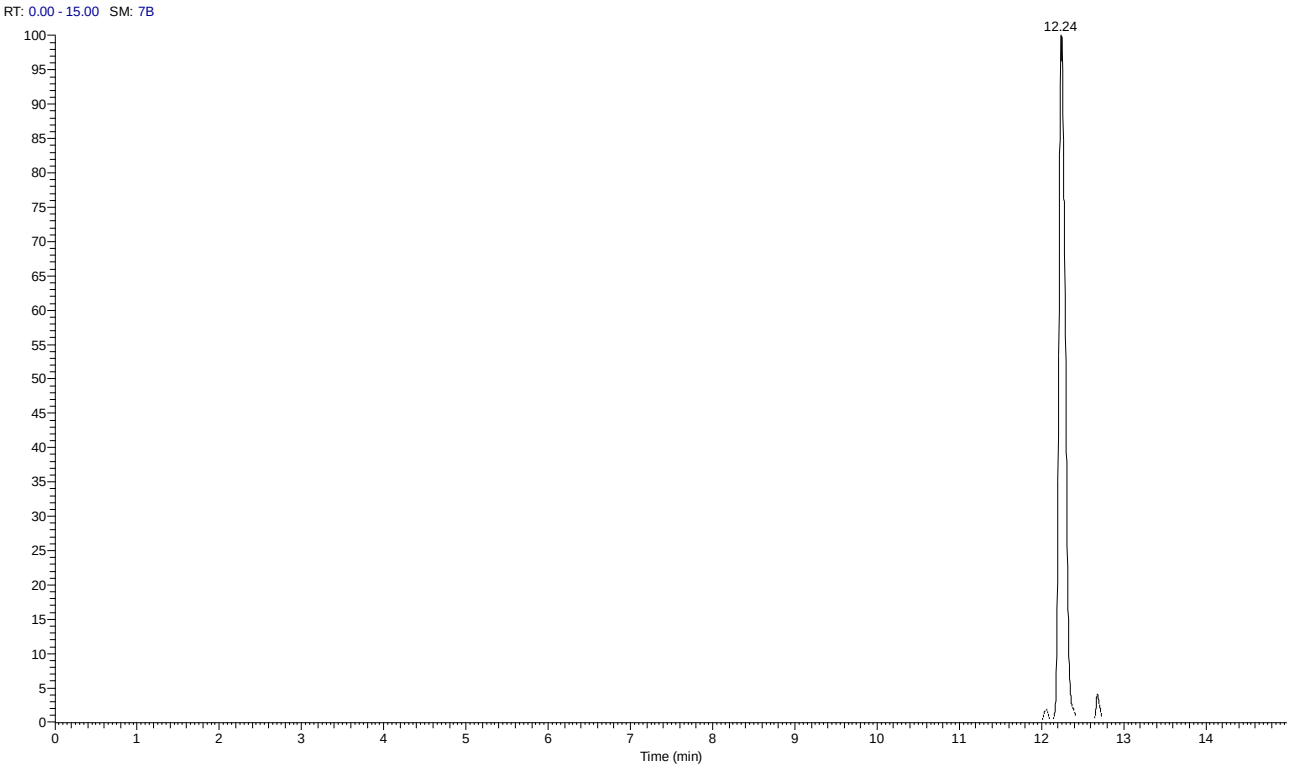


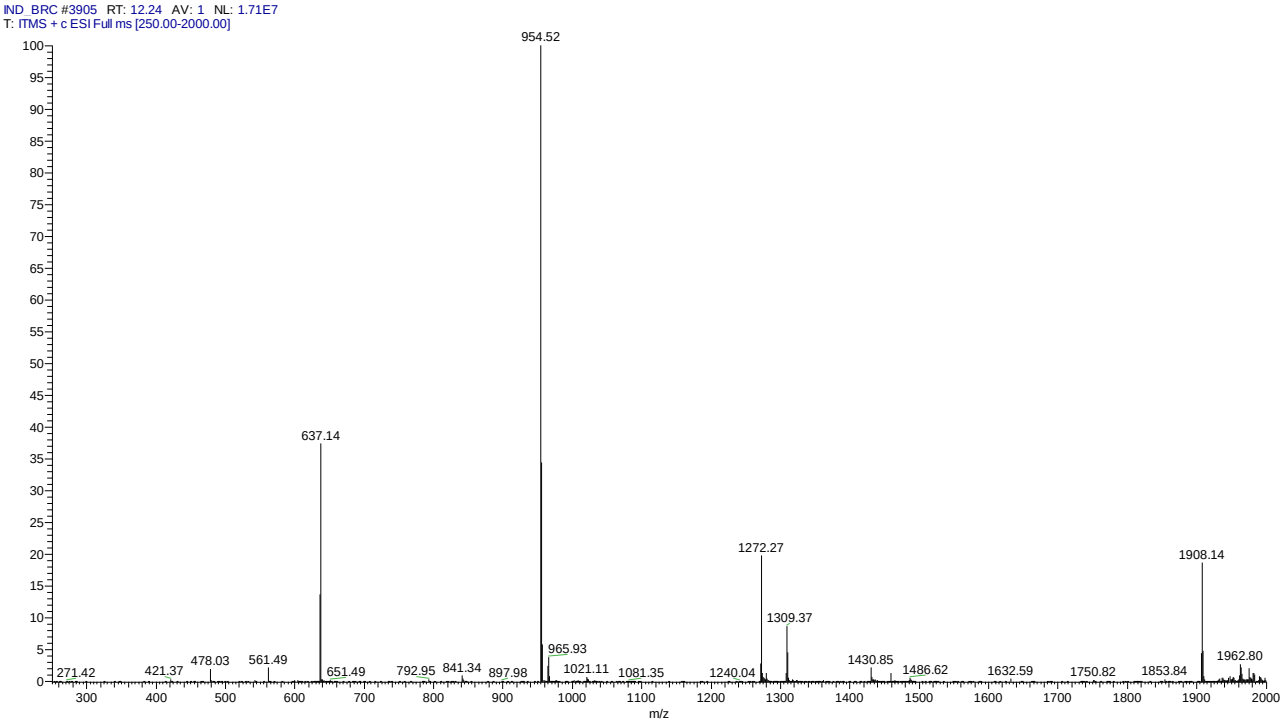
CP1. MW: 3338.9. Expected m/z values: $[M+2H]^{2+}$: 1670.46; $[M+3H]^{3+}$: 1113.97 $[M+4H]^{4+}$: 835.73



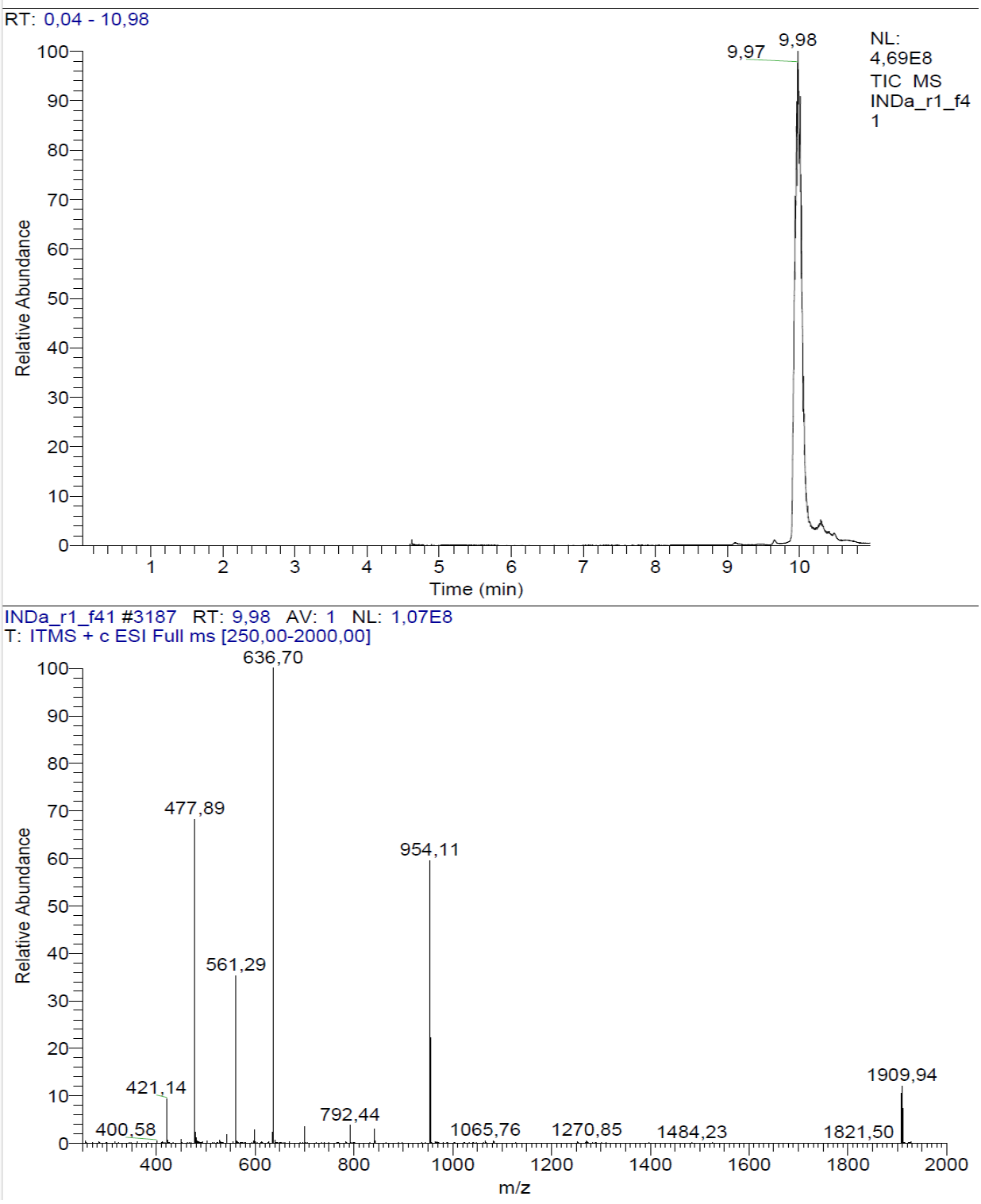


GUA2. MW: 2643.05. Expected m/z values: $[M+2H]^{2+}$: 1322.53; $[M+3H]^{3+}$: 882.02; $[M+4H]^{4+}$: 661.77

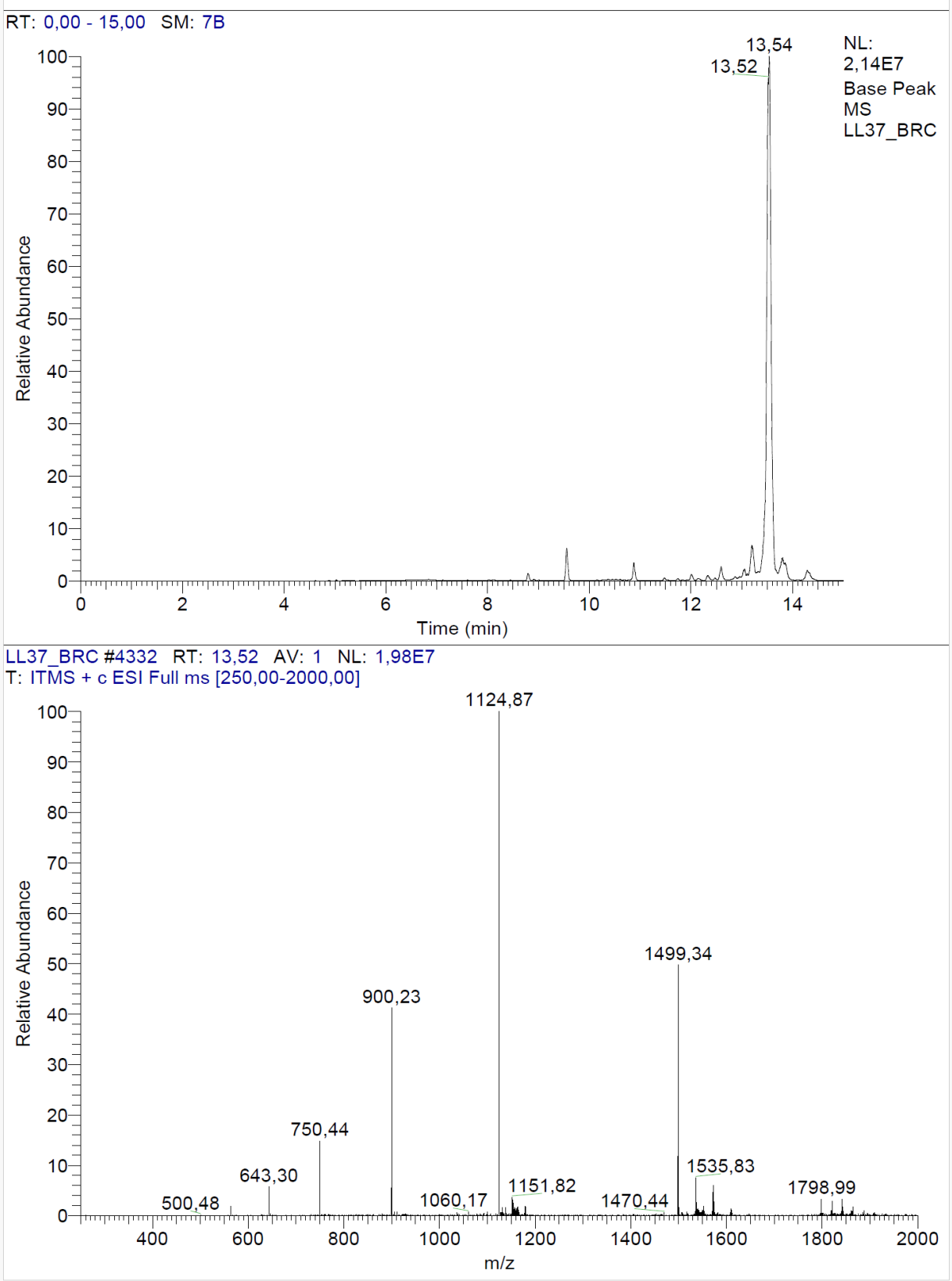




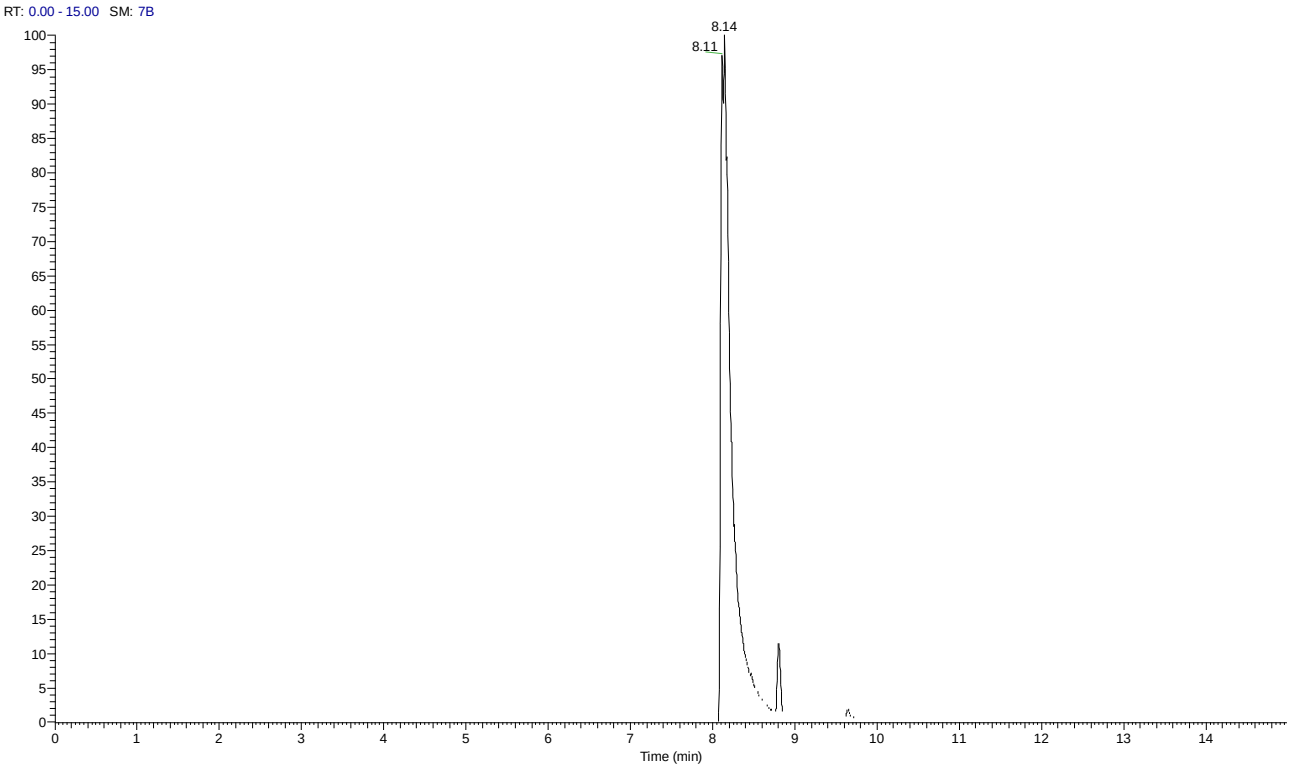
IND. MW: 1907.29. Expected m/z values: $[M+H]^{+}$:1908.30; $[M+2H]^{2+}$: 954.65; $[M+3H]^{3+}$: 636.77 $[M+4H]^{4+}$: 477.83

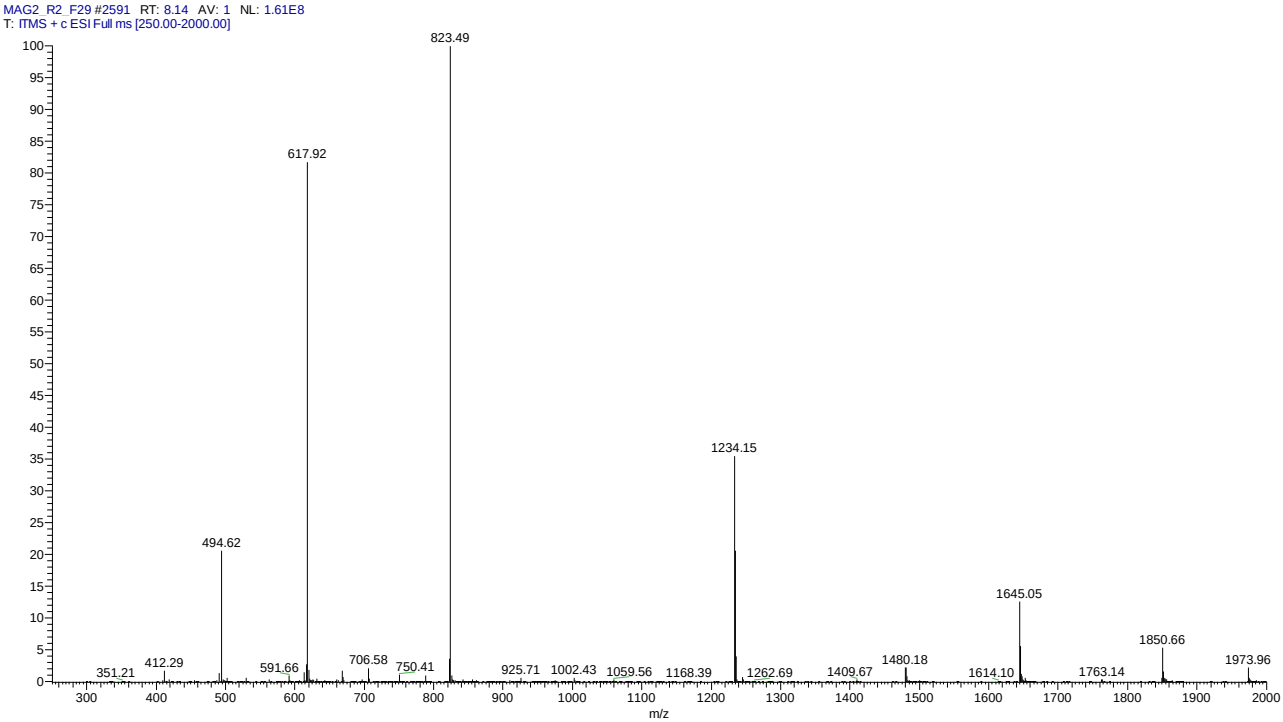


INDa. MW: 1906.29. Expected m/z values: $[M+H]^{+}$:1907.30; $[M+2H]^{2+}$: 954.15; $[M+3H]^{3+}$: 636.44 $[M+4H]^{4+}$: 477.58

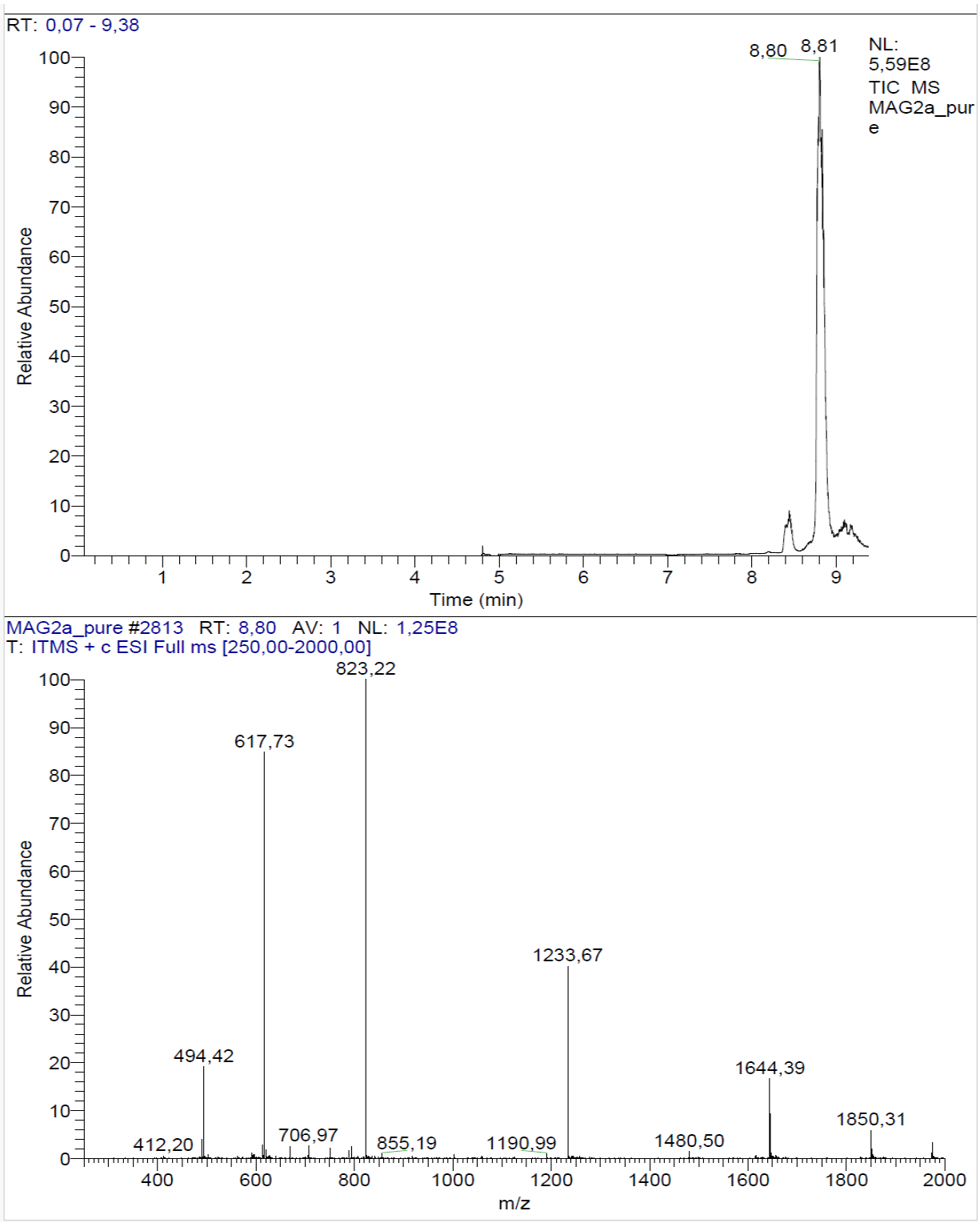


LL37. MW: 4493.34. Expected m/z values: $[M+3H]^{3+}$: 1498.79 $[M+4H]^{4+}$: 1124.34

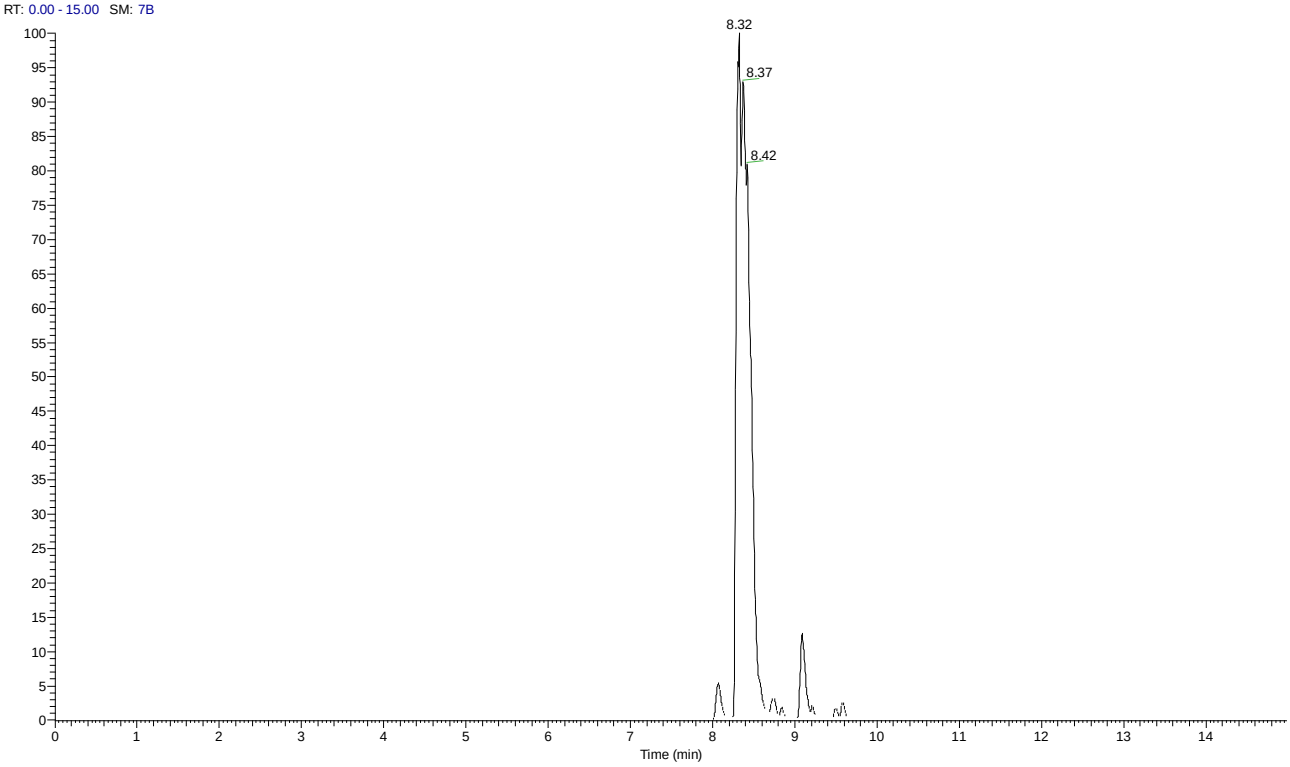


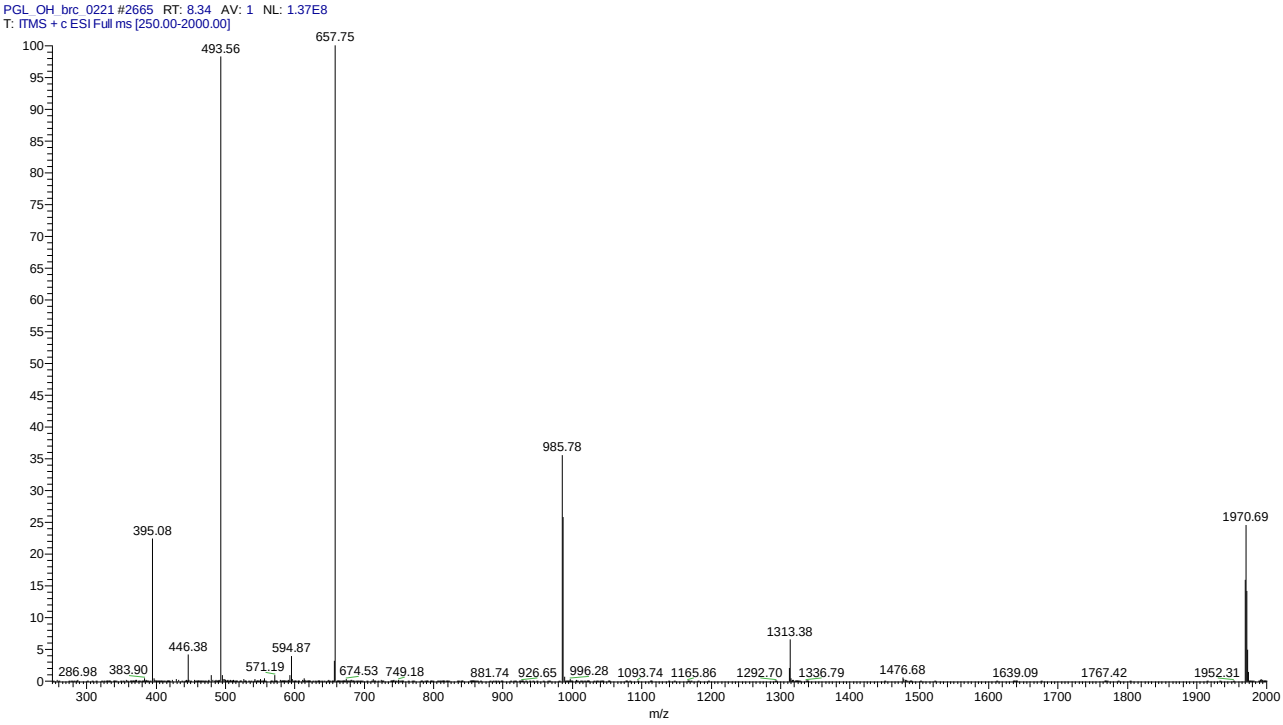


MAG2. MW: 2466.9. Expected m/z values: $[M+2H]^{2+}$: 1234.46; $[M+3H]^{3+}$: 823.31 $[M+4H]^{4+}$: 617.73

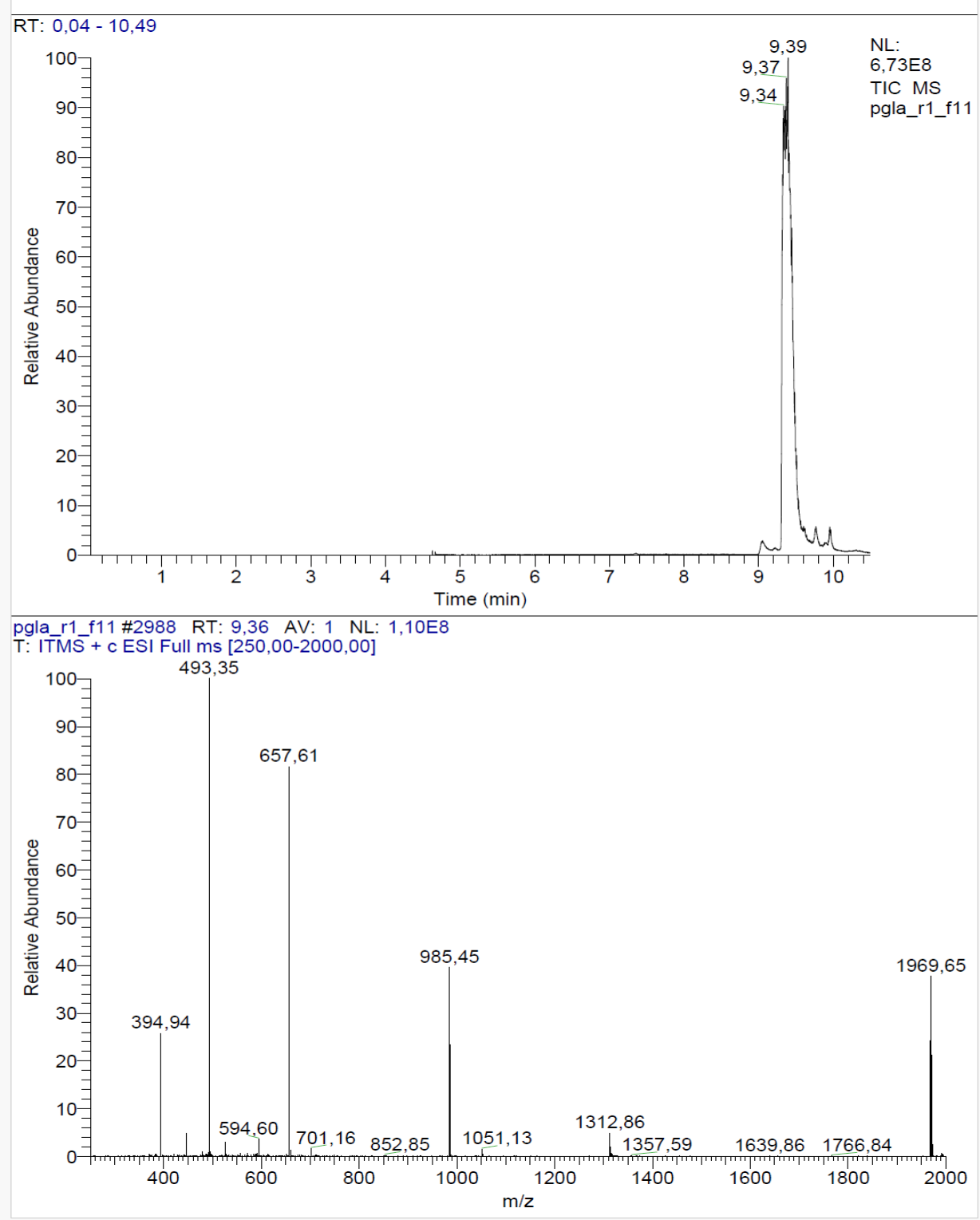


MAG2a. MW: 2465.9. Expected m/z values: $[M+2H]^{2+}$: 1233.96; $[M+3H]^{3+}$: 822.97 $[M+4H]^{4+}$: 617.48

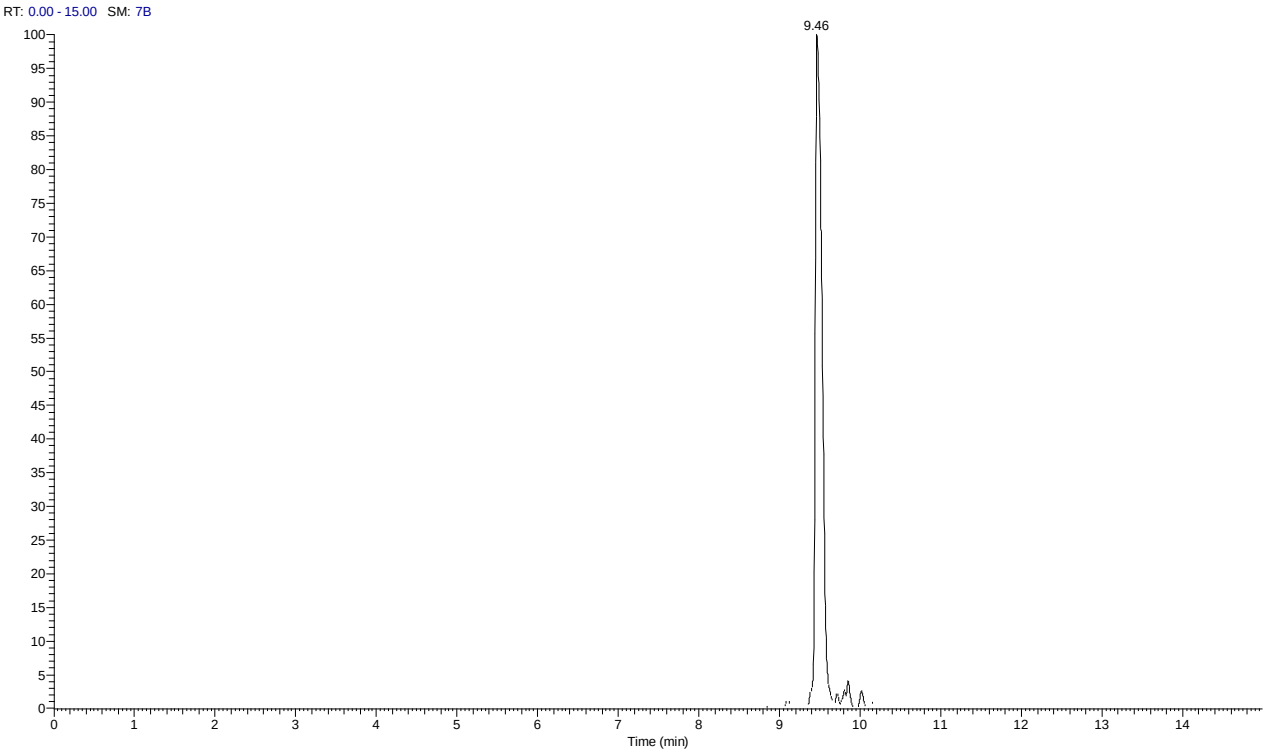


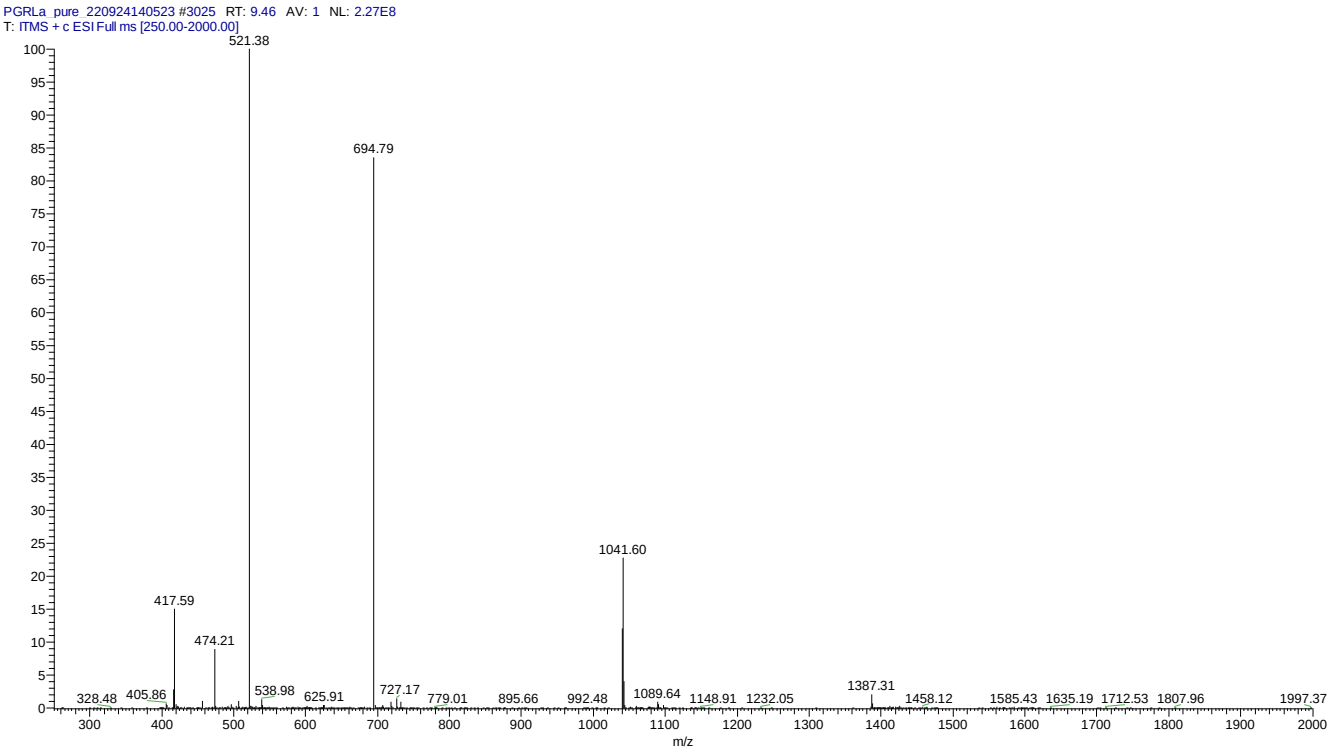


PGL. MW: 1969.4. Expected m/z values: $[M+H]^{+}$:1970.46; $[M+2H]^{2+}$: 985.73; $[M+3H]^{3+}$: 657.49 $[M+4H]^{4+}$: 493.37

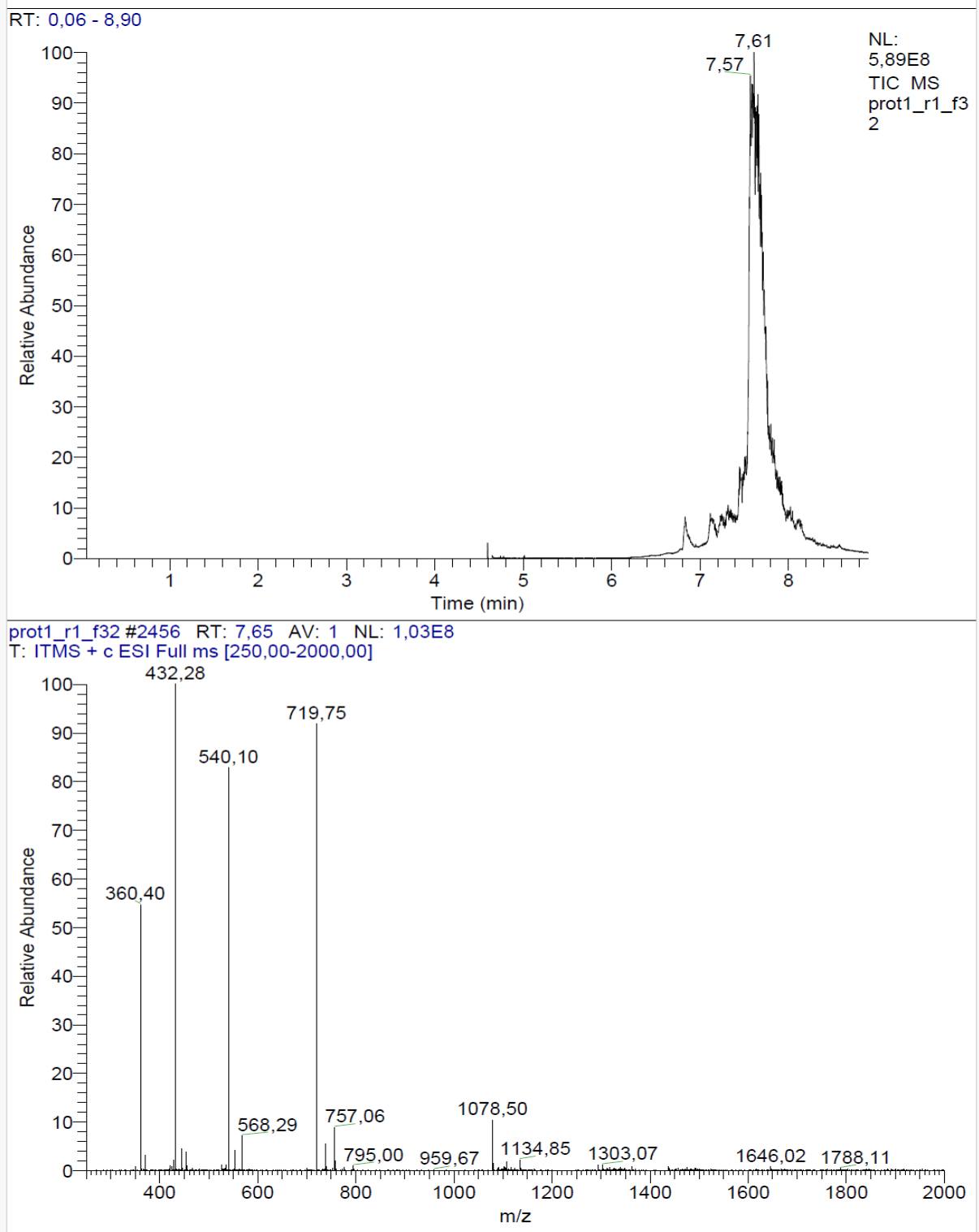


PGLa. MW: 1968.4. Expected m/z values: $[M+H]^{+}$:1969.4; $[M+2H]^{2+}$: 985.2; $[M+3H]^{3+}$: 657.1 $[M+4H]^{4+}$: 493.1

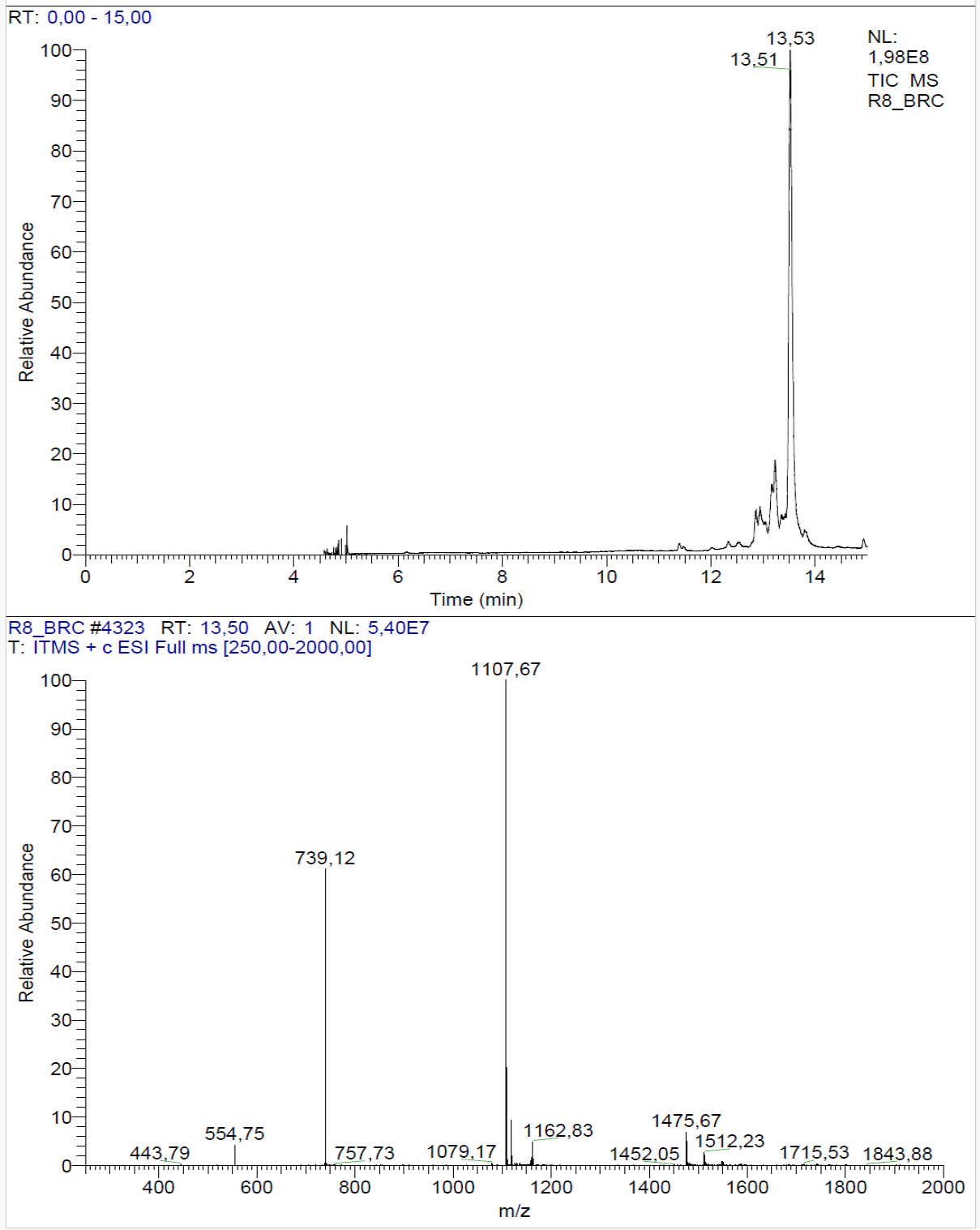




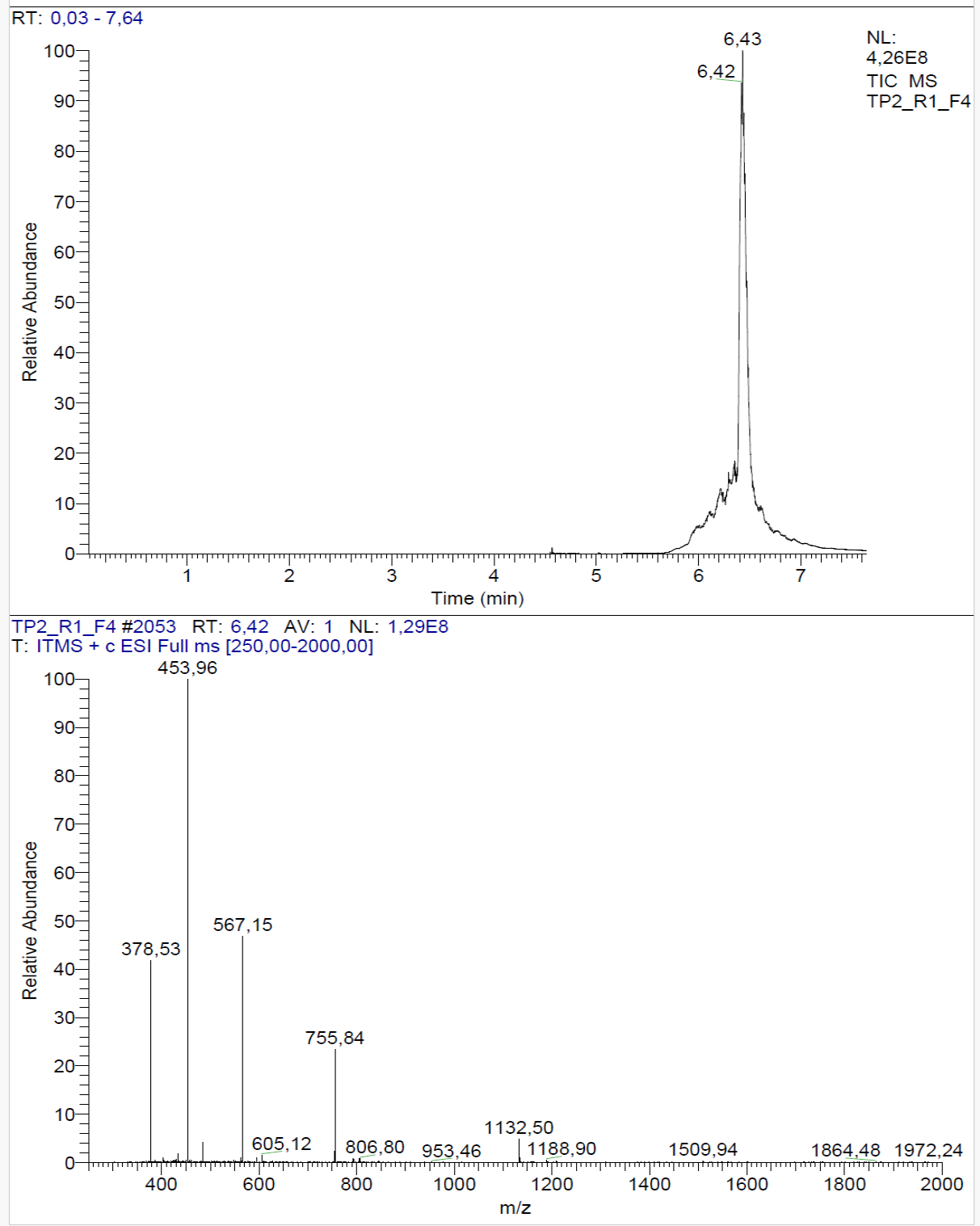
PGRLa. MW: 2080.51. Expected m/z values: $[M+2H]^{2+}$: 1041.26; $[M+3H]^{3+}$: 694.51 $[M+4H]^{4+}$: 521.13



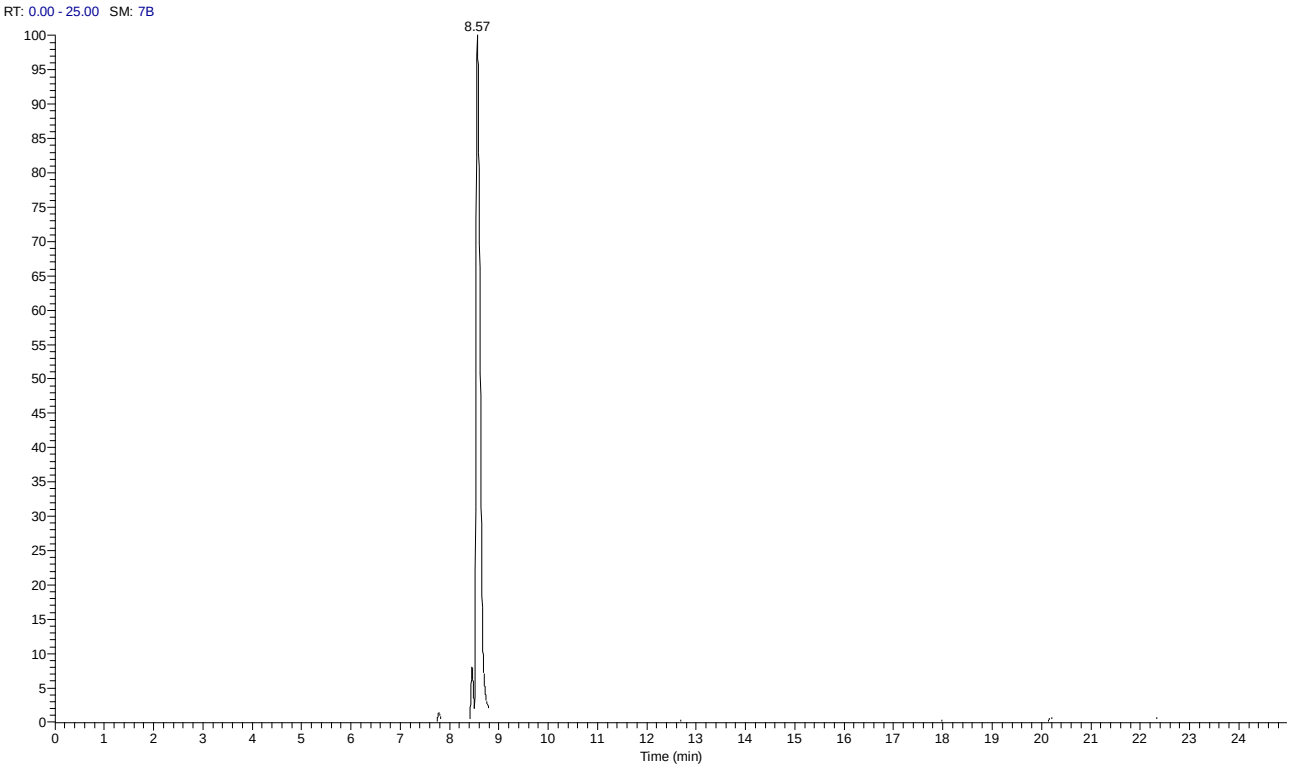
PROT1 (oxidized). MW: 2155.68. Expected m/z values: $[M+2H]^{2+}$: 1078.85; $[M+3H]^{3+}$: 719.57 $[M+4H]^{4+}$: 539.93

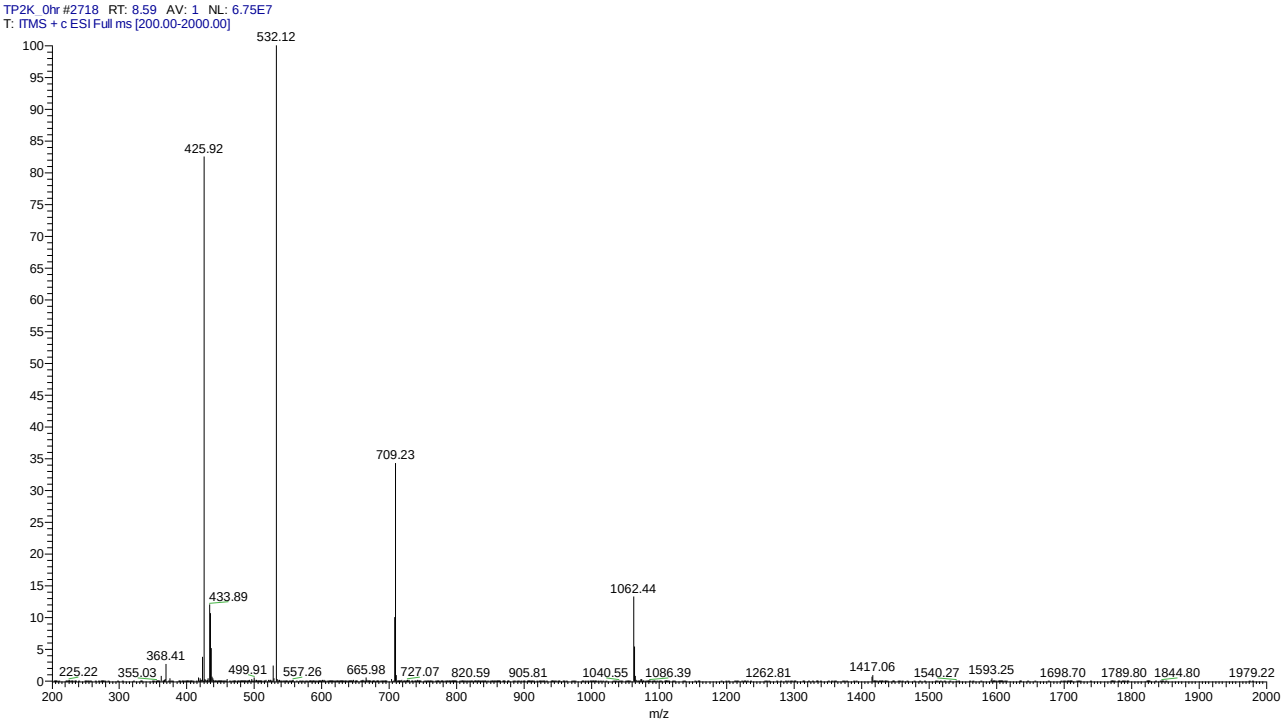


R8. MW: 2213.76. Expected m/z values: $[M+2H]^{2+}$: 1107.89; $[M+3H]^{3+}$: 738.93 $[M+4H]^{4+}$: 554.45



TP2 (oxidized). MW: 2263.82. Expected m/z values: $[M+2H]^{2+}$: 1132.92; $[M+3H]^{3+}$: 755.61 $[M+4H]^{4+}$: 566.96





TP2Ka (oxidized). MW: 2123.72. Expected m/z values: $[M+2H]^{2+}$: 1062.87; $[M+3H]^{3+}$: 708.91 $[M+4H]^{4+}$: 531.94